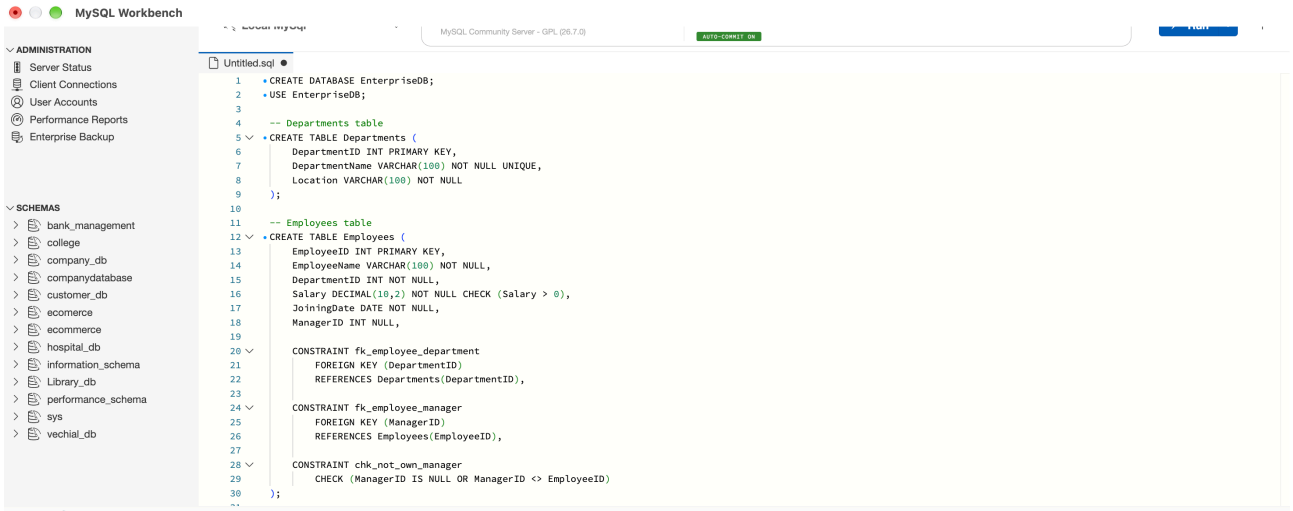
Name : Gorlamandala Nanda Sai
USN : 23BTRCL042
CSE-AIML
HCLID - HCLTFP2262836

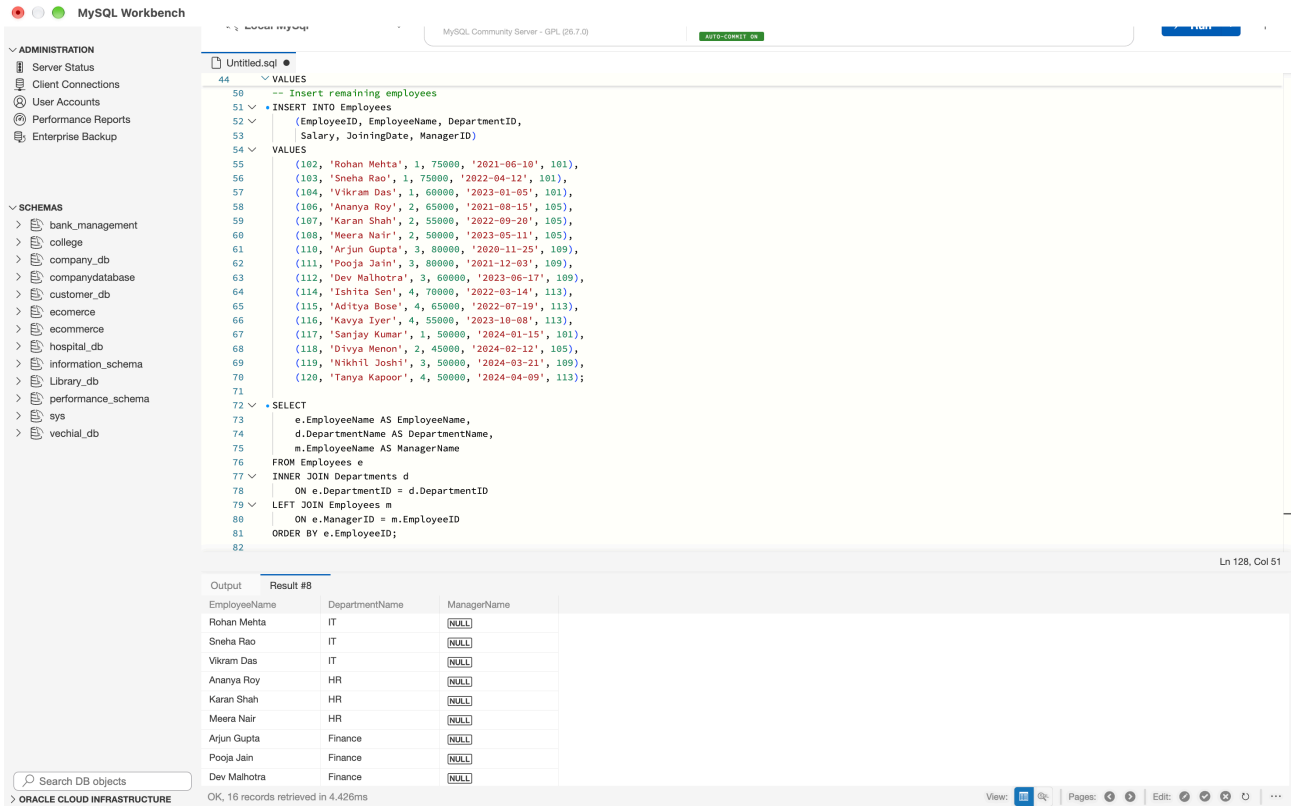
Q-1: Database Design and Advanced Queries

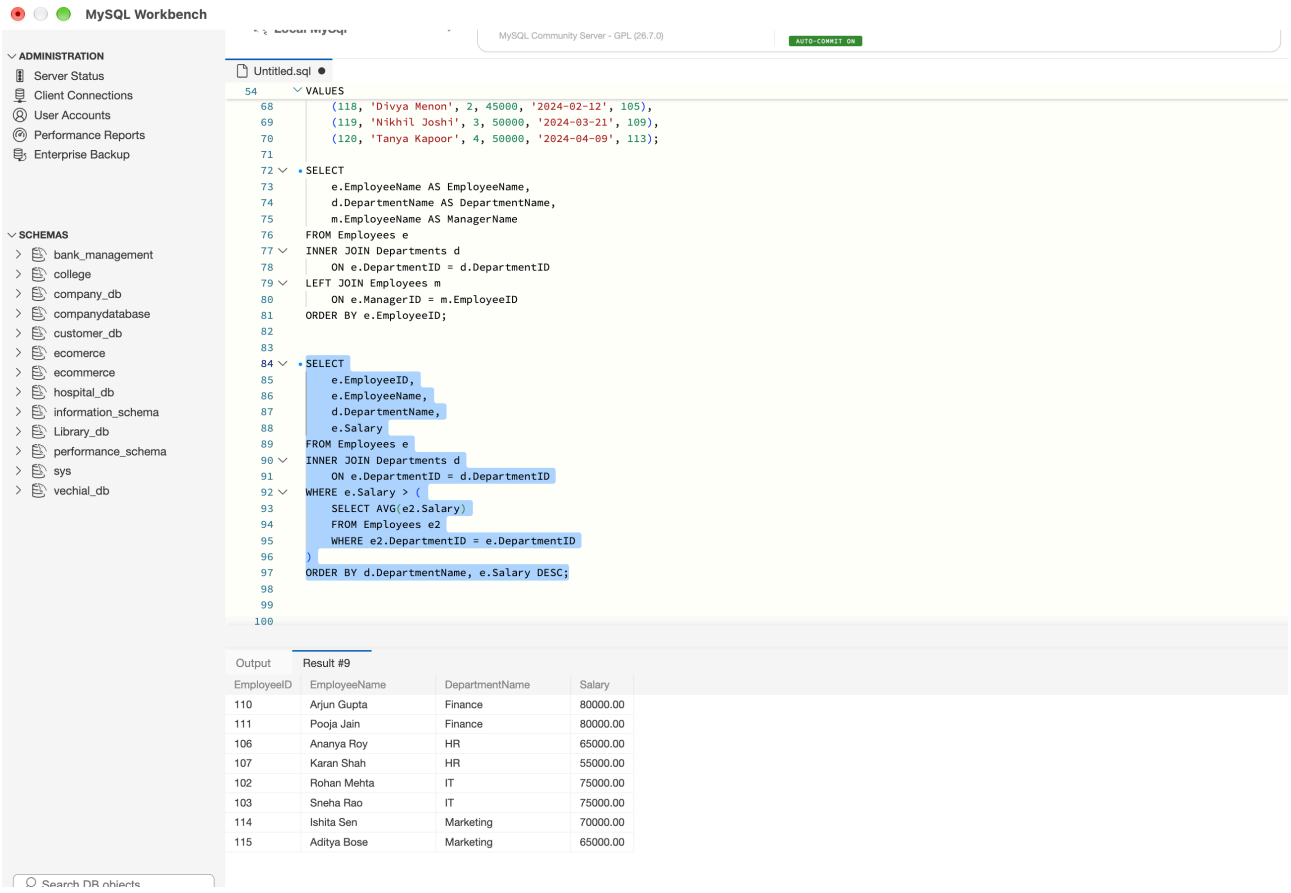


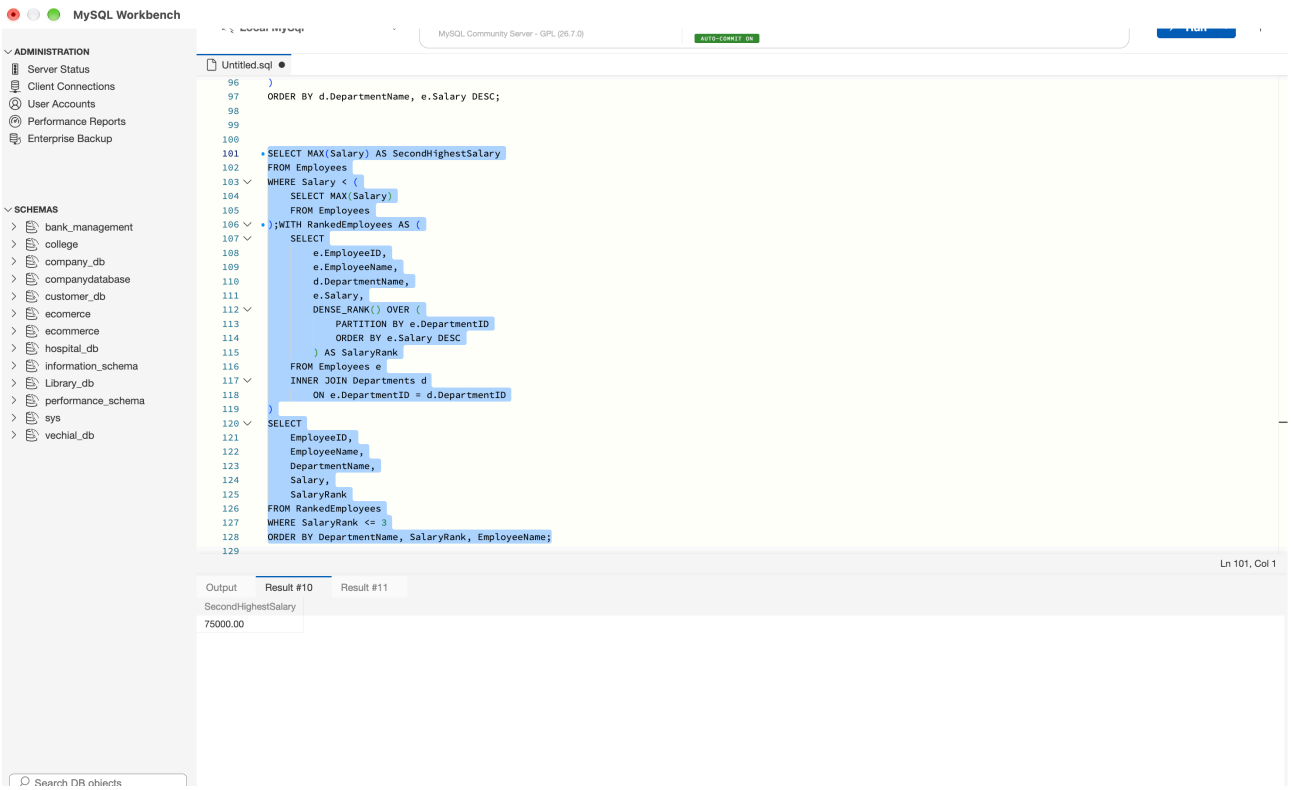

```sql
INSERT INTO Departments
    (DepartmentID, DepartmentName, Location)
VALUES
    (1, 'IT', 'Bengaluru'),
    (2, 'HR', 'Mumbai'),
    (3, 'Finance', 'Delhi'),
    (4, 'Marketing', 'Hyderabad'),
    (5, 'Research', 'Pune');

INSERT INTO Employees
    (EmployeeID, EmployeeName, DepartmentID,
     Salary, JoiningDate, ManagerID)
VALUES
    (101, 'Amit Sharma', 1, 90000, '2020-01-15', NULL),
    (105, 'Priya Singh', 2, 85000, '2019-03-10', NULL),
    (109, 'Rahul Verma', 3, 95000, '2018-07-20', NULL),
    (113, 'Neha Patel', 4, 80000, '2021-02-01', NULL);
```

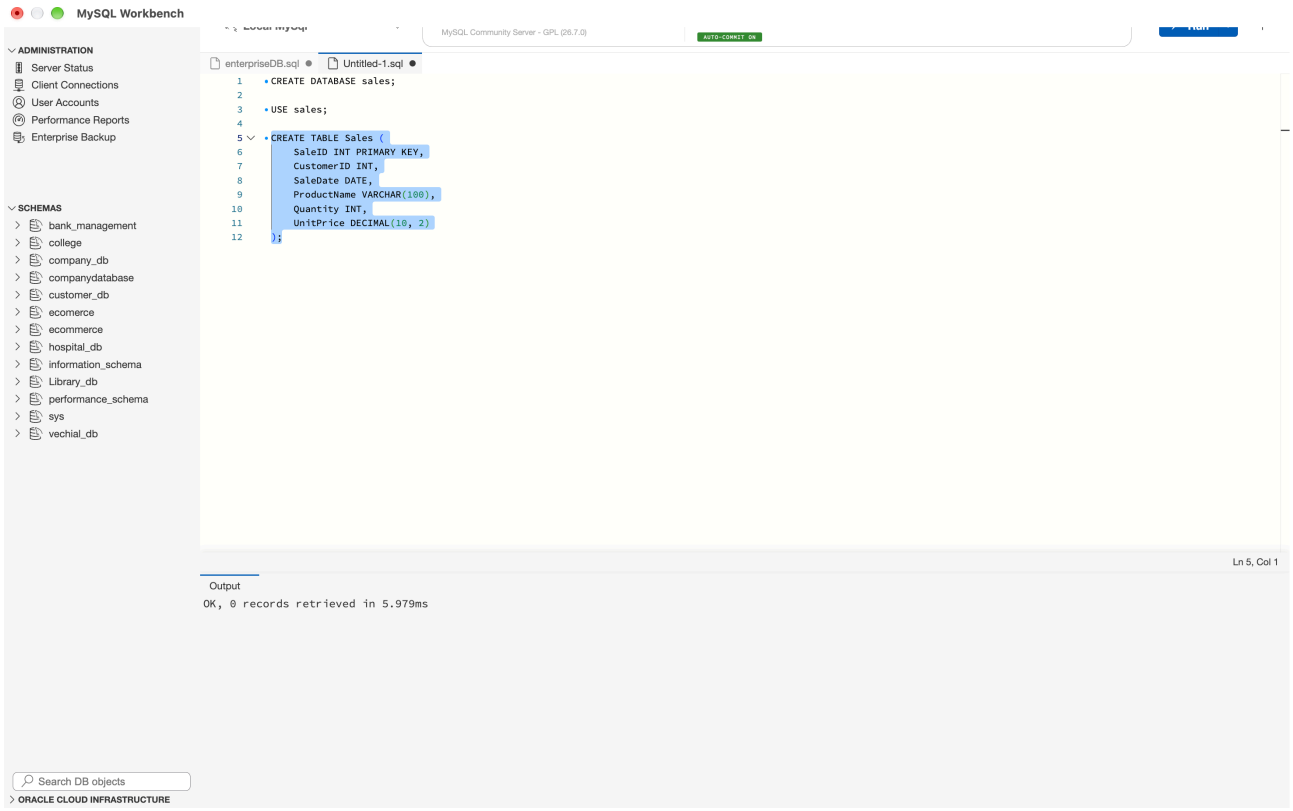
```sql
-- Insert remaining employees
INSERT INTO Employees
    (EmployeeID, EmployeeName, DepartmentID,
     Salary, JoiningDate, ManagerID)
VALUES
    (102, 'Rohan Mehta', 1, 75000, '2021-06-10', 101),
    (103, 'Sneha Rao', 1, 75000, '2022-04-12', 101),
    (104, 'Vikram Das', 1, 60000, '2023-01-05', 101),
    (106, 'Ananya Roy', 2, 65000, '2021-08-15', 105),
    (107, 'Karan Shah', 2, 55000, '2022-09-20', 105),
    (108, 'Meera Nair', 2, 50000, '2023-05-11', 105),
    (110, 'Arjun Gupta', 3, 80000, '2020-11-25', 109),
    (111, 'Pooja Jain', 3, 80000, '2021-12-03', 109),
    (112, 'Dev Malhotra', 3, 60000, '2023-06-17', 109),
    (114, 'Ishita Sen', 4, 70000, '2022-03-14', 113),
    (115, 'Aditya Bose', 4, 65000, '2022-07-19', 113),
    (116, 'Kavya Iyer', 4, 55000, '2023-10-08', 113),
    (117, 'Sanjay Kumar', 1, 50000, '2024-01-15', 101),
    (118, 'Divya Menon', 2, 45000, '2024-02-12', 105),
    (119, 'Nikhil Joshi', 3, 50000, '2024-03-21', 109),
    (120, 'Tanya Kapoor', 4, 50000, '2024-04-09', 113);
```

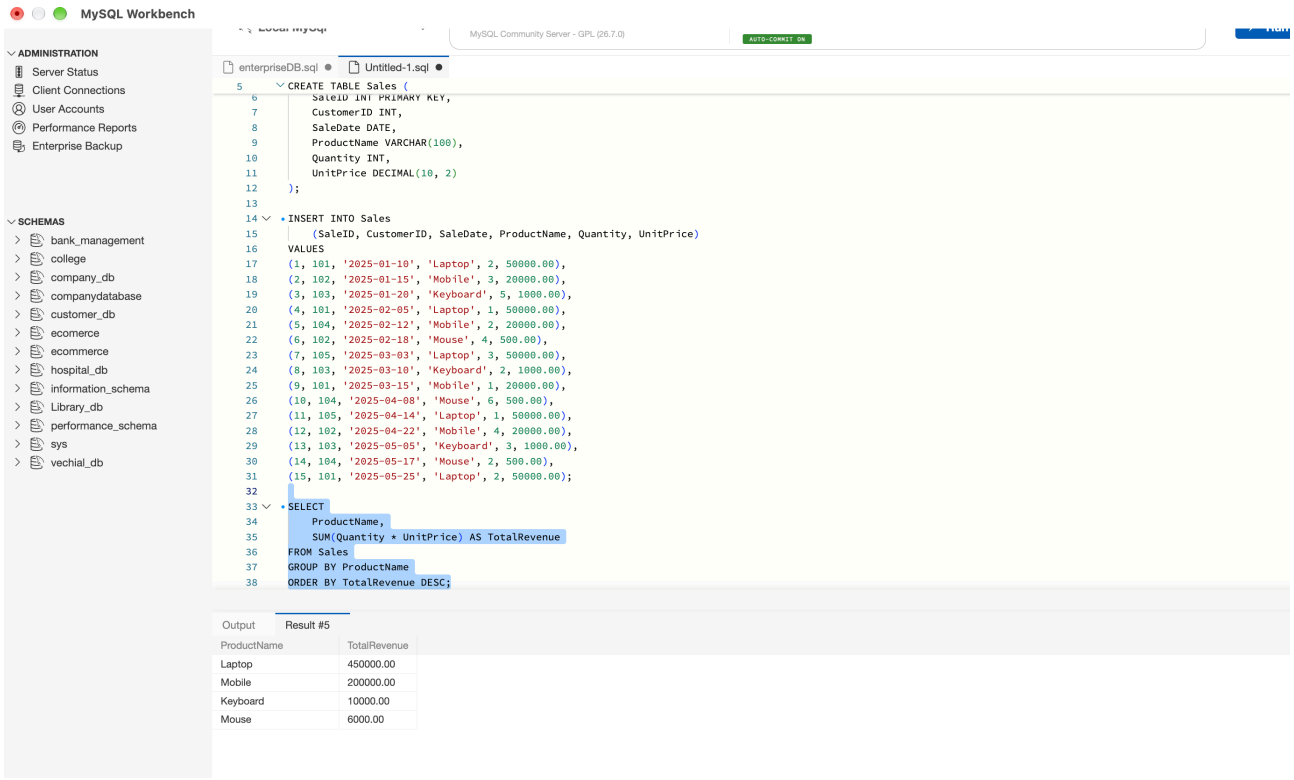

MySQL Workbench
ADMINISTRATION
Server Status
Client Connections
User Accounts
Performance Reports
Enterprise Backup
SCHEMAS
bank_management
college
company_db
companydatabase
customer_db
ecomerce
ecommerce
hospital_db
information_schema
Library_db
performance_schema
sys
vechial_db
Search DB objects
ORACLE CLOUD INFRASTRUCTURE
MySQL Community Server - GPL (26.7.0)
AUTO-COMMIT ON
Untitled.sql
VALUES
-- Insert remaining employees
INSERT INTO Employees
(EmployeeID, EmployeeName, DepartmentID,
Salary, JoiningDate, ManagerID)
VALUES
(102, 'Rohan Mehta', 1, 75000, '2021-06-10', 101),
(103, 'Sneha Rao', 1, 75000, '2022-04-12', 101),
(104, 'Vikram Das', 1, 60000, '2023-01-05', 101),
(106, 'Ananya Roy', 2, 65000, '2021-08-15', 105),
(107, 'Karan Shah', 2, 55000, '2022-09-20', 105),
(108, 'Meera Nair', 2, 50000, '2023-05-11', 105),
(110, 'Arjun Gupta', 3, 80000, '2020-11-25', 109),
(111, 'Pooja Jain', 3, 80000, '2021-12-03', 109),
(112, 'Dev Malhotra', 3, 60000, '2023-06-17', 109),
(114, 'Ishita Sen', 4, 70000, '2022-03-14', 113),
(115, 'Aditya Bose', 4, 65000, '2022-07-19', 113),
(116, 'Kavya Iyer', 4, 55000, '2023-10-08', 113),
(117, 'Sanjay Kumar', 1, 50000, '2024-01-15', 101),
(118, 'Divya Menon', 2, 45000, '2024-02-12', 105),
(119, 'Nikhil Joshi', 3, 50000, '2024-03-21', 109),
(120, 'Tanya Kapoor', 4, 50000, '2024-04-09', 113);
SELECT
e.EmployeeName AS EmployeeName,
d.DepartmentName AS DepartmentName,
m.EmployeeName AS ManagerName
FROM Employees e
INNER JOIN Departments d
ON e.DepartmentID = d.DepartmentID
LEFT JOIN Employees m
ON e.ManagerID = m.EmployeeID
ORDER BY e.EmployeeID;
Ln 128, Col 51
Output
Result #8
EmployeeName
DepartmentName
ManagerName
Rohan Mehta
IT
NULL
Sneha Rao
IT
NULL
Vikram Das
IT
NULL
Ananya Roy
HR
NULL
Karan Shah
HR
NULL
Meera Nair
HR
NULL
Arjun Gupta
Finance
NULL
Pooja Jain
Finance
NULL
Dev Malhotra
Finance
NULL
OK, 16 records retrieved in 4.426ms
View:
Pages:
Edit:


MySQL Workbench
ADMINISTRATION
Server Status
Client Connections
User Accounts
Performance Reports
Enterprise Backup
SCHEMAS
bank_management
college
company_db
companydatabase
customer_db
ecomerce
ecommerce
hospital_db
information_schema
Library_db
performance_schema
sys
vechial_db
Search DB objects
MySQL Community Server - GPL (26.7.0)
AUTO-COMMIT ON
Untitled.sql
VALUES
(118, 'Divya Menon', 2, 45000, '2024-02-12', 105),
(119, 'Nikhil Joshi', 3, 50000, '2024-03-21', 109),
(120, 'Tanya Kapoor', 4, 50000, '2024-04-09', 113);
SELECT
e.EmployeeName AS EmployeeName,
d.DepartmentName AS DepartmentName,
m.EmployeeName AS ManagerName
FROM Employees e
INNER JOIN Departments d
ON e.DepartmentID = d.DepartmentID
LEFT JOIN Employees m
ON e.ManagerID = m.EmployeeID
ORDER BY e.EmployeeID;
SELECT
e.EmployeeID,
e.EmployeeName,
d.DepartmentName,
e.Salary
FROM Employees e
INNER JOIN Departments d
ON e.DepartmentID = d.DepartmentID
WHERE e.Salary > (
SELECT AVG(e2.Salary)
FROM Employees e2
WHERE e2.DepartmentID = e.DepartmentID
)
ORDER BY d.DepartmentName, e.Salary DESC;
Output
Result #9
EmployeeID
EmployeeName
DepartmentName
Salary
110
Arjun Gupta
Finance
80000.00
111
Pooja Jain
Finance
80000.00
106
Ananya Roy
HR
65000.00
107
Karan Shah
HR
55000.00
102
Rohan Mehta
IT
75000.00
103
Sneha Rao
IT
75000.00
114
Ishita Sen
Marketing
70000.00
115
Aditya Bose
Marketing
65000.00


MySQL Workbench
MySQL Community Server - GPL (26.7.0)
AUTO-COMMIT ON
ADMINISTRATION
Server Status
Client Connections
User Accounts
Performance Reports
Enterprise Backup
SCHEMAS
bank_management
college
company_db
companydatabase
customer_db
ecomerce
ecommerce
hospital_db
information_schema
Library_db
performance_schema
sys
vechial_db
Search DB objects
Untitled.sql
)
ORDER BY d.DepartmentName, e.Salary DESC;
SELECT MAX(Salary) AS SecondHighestSalary
FROM Employees
WHERE Salary < (
SELECT MAX(Salary)
FROM Employees
);WITH RankedEmployees AS (
SELECT
e.EmployeeID,
e.EmployeeName,
d.DepartmentName,
e.Salary,
DENSE_RANK() OVER (
PARTITION BY e.DepartmentID
ORDER BY e.Salary DESC
) AS SalaryRank
FROM Employees e
INNER JOIN Departments d
ON e.DepartmentID = d.DepartmentID
)
SELECT
EmployeeID,
EmployeeName,
DepartmentName,
Salary,
SalaryRank
FROM RankedEmployees
WHERE SalaryRank <= 3
ORDER BY DepartmentName, SalaryRank, EmployeeName;
Ln 101, Col 1
Output
Result #10
Result #11
SecondHighestSalary
75000.00

Q-2: Sales Analytics and Query Optimization



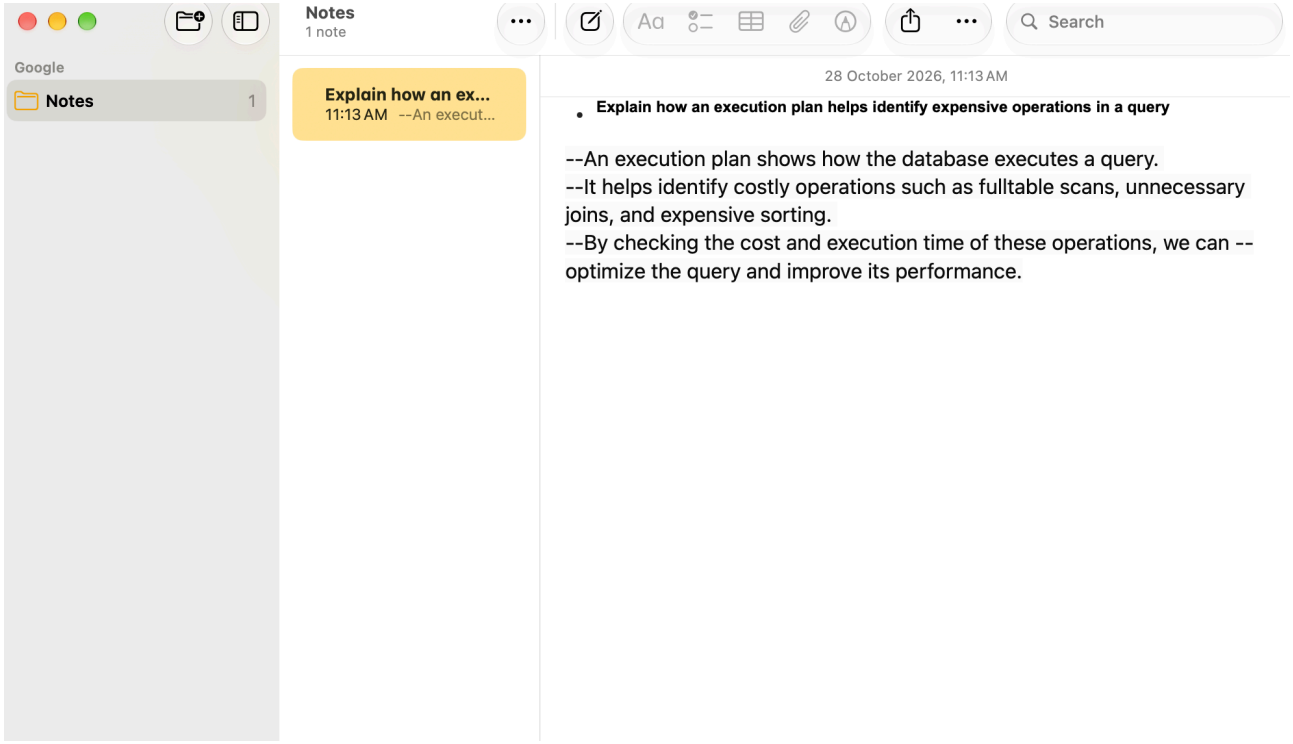


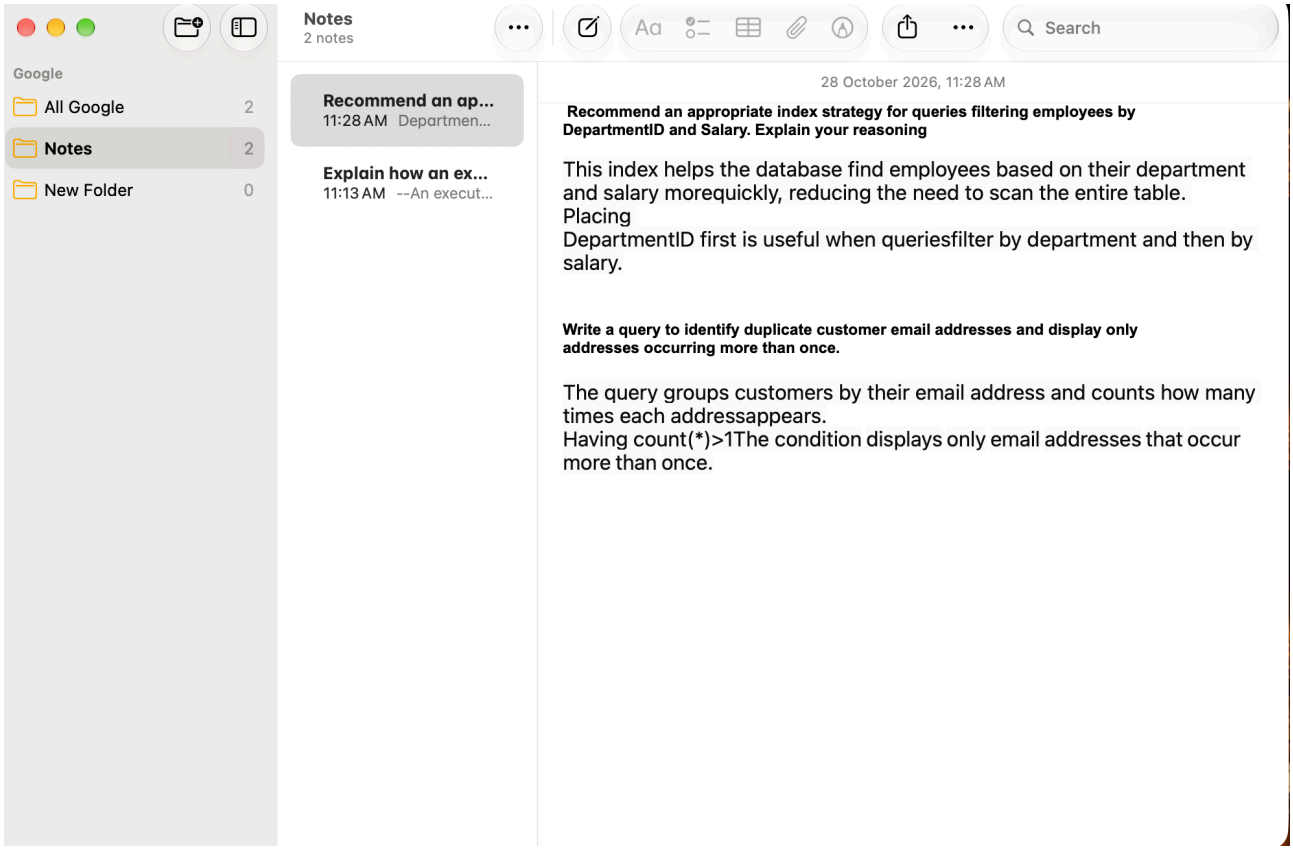
ProductName	TotalRevenue
Laptop	450000.00
Mobile	200000.00
Keyboard	10000.00
Mouse	6000.00



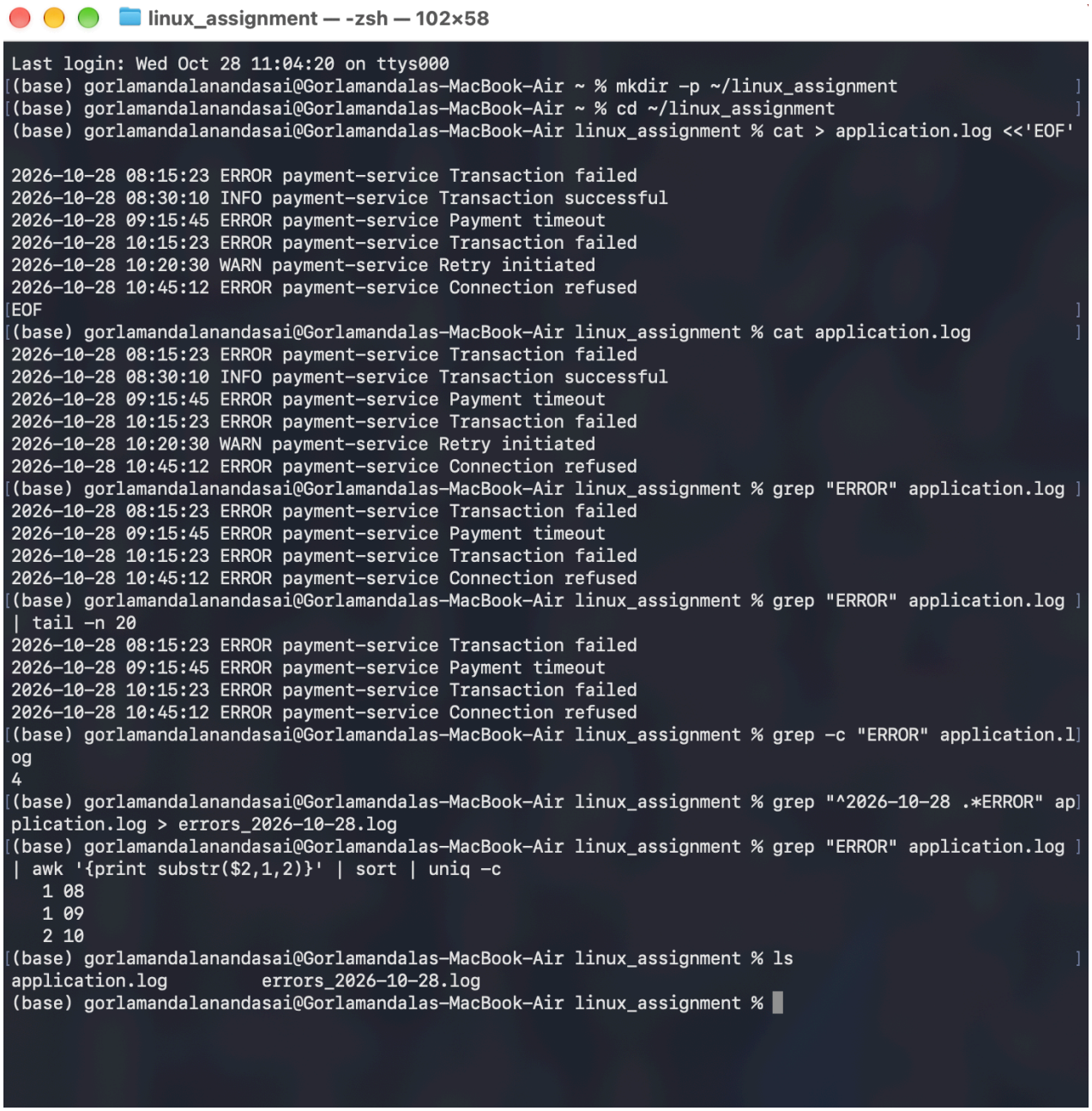

```sql
SELECT *
FROM Employees
WHERE JoiningDate >= '2025-01-01'
  AND JoiningDate <  '2026-01-01';
```

Q-3: SQL Troubleshooting

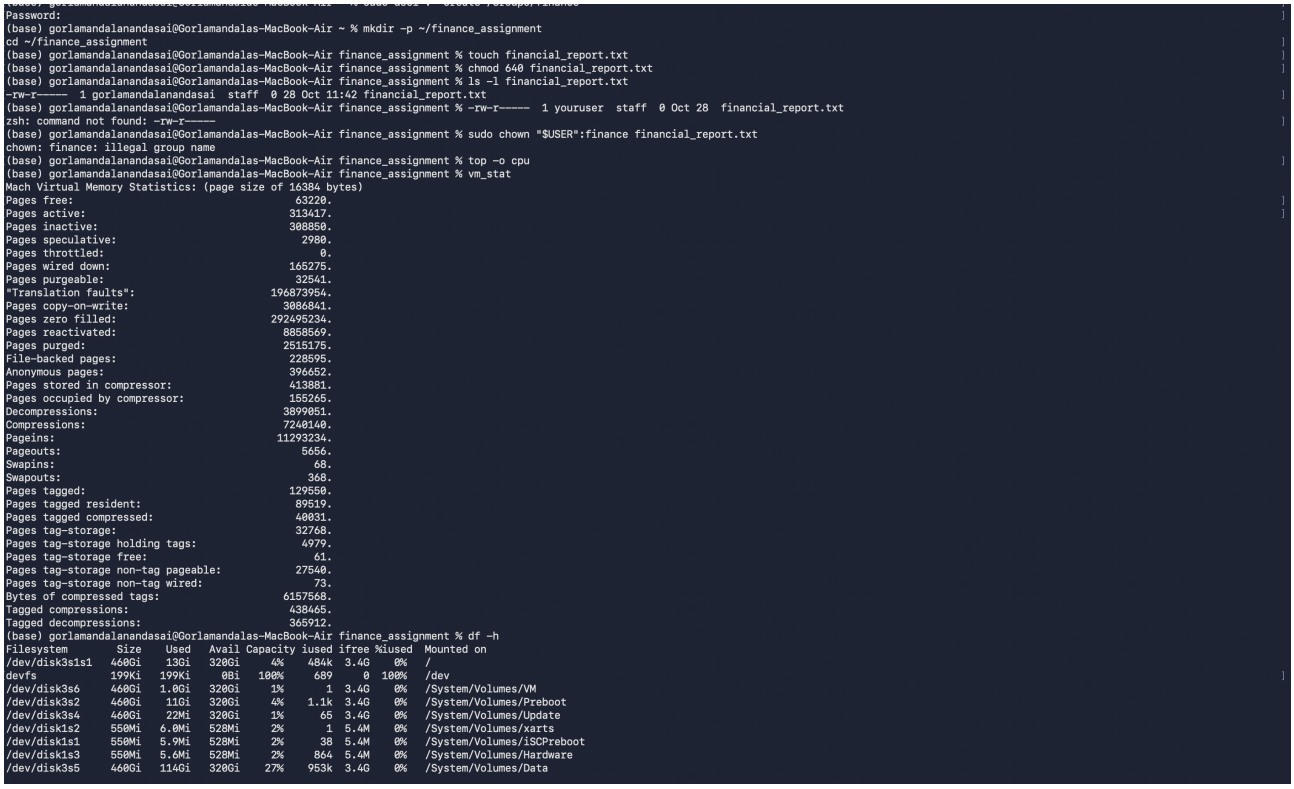




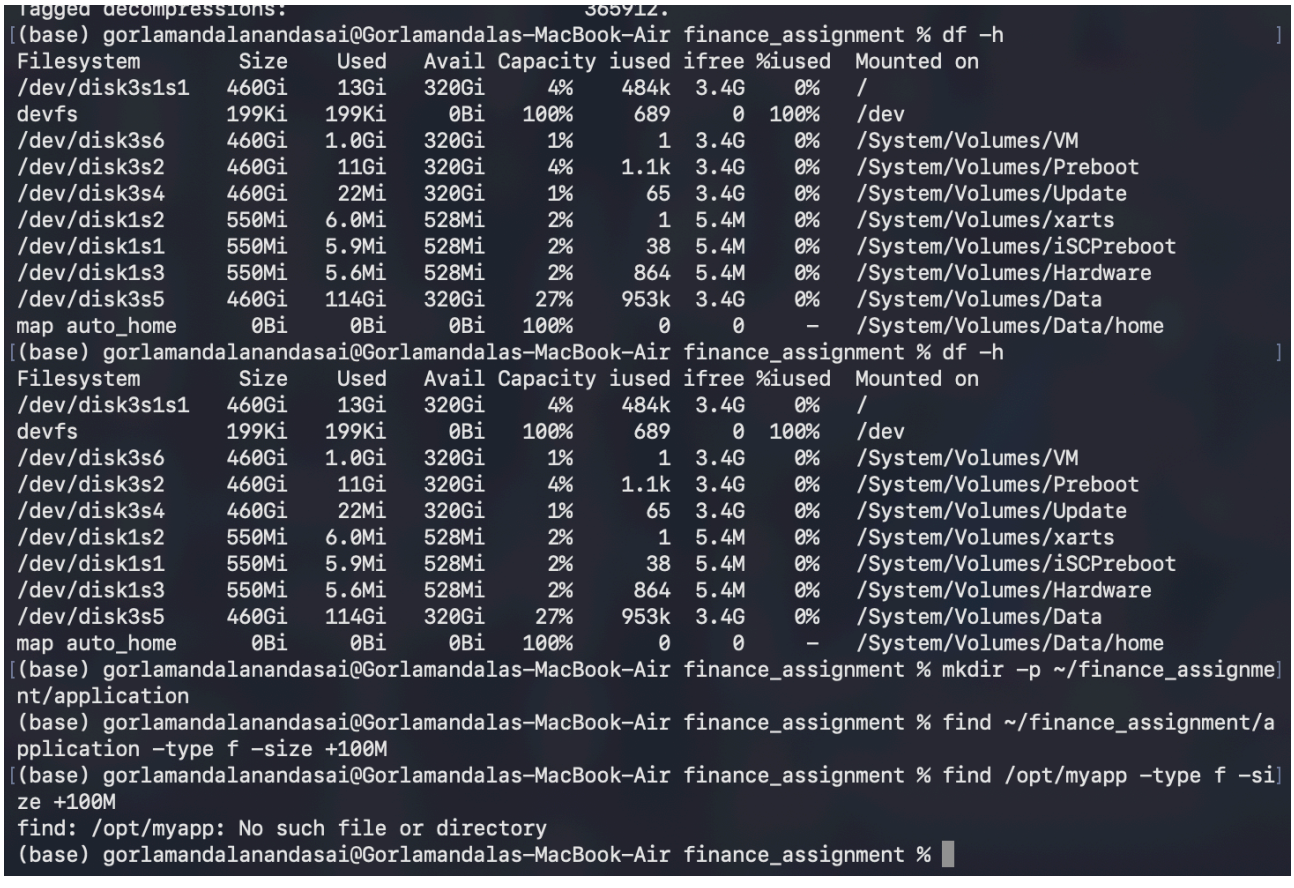
Question 4: Log Analysis



Question 5: Permissions and System Troubleshooting:

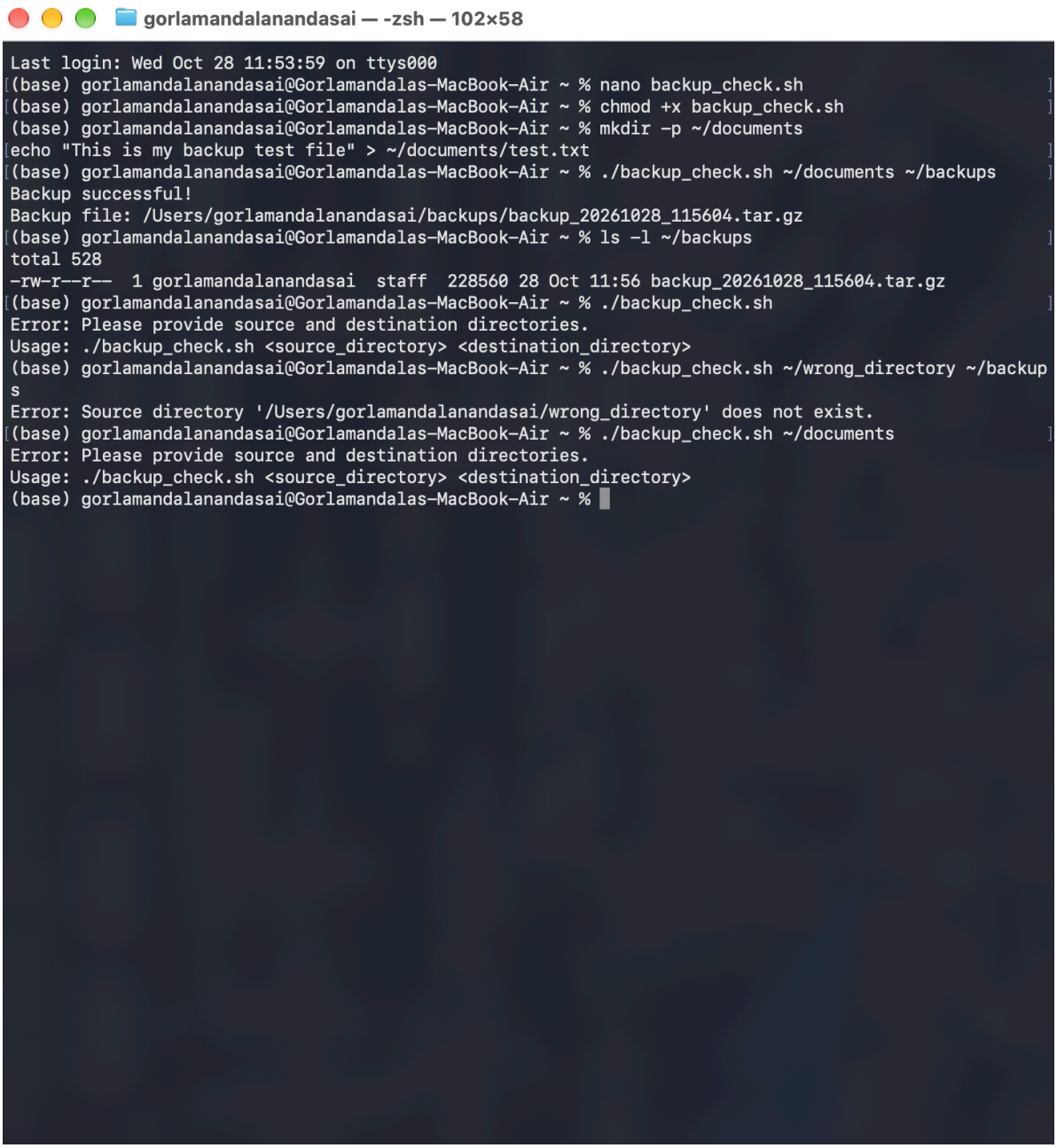


```
Password:
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air ~ % mkdir -p ~/finance_assignment
cd ~/finance_assignment
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % touch financial_report.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % chmod 640 financial_report.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % ls -l financial_report.txt
-rw-r-----  1 gorlamandalanandasai  staff  0 28 Oct 11:42 financial_report.txt
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % -rw-r-----  1 youruser  staff  0 Oct 28  financial_report.txt
zsh: command not found: -rw-r-----
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % sudo chown "$USER":finance financial_report.txt
chown: finance: illegal group name
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % top -o cpu
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % vm_stat
Mach Virtual Memory Statistics: (page size of 16384 bytes)
Pages free:                                63220.
Pages active:                             313417.
Pages inactive:                           308850.
Pages speculative:                          2980.
Pages throttled:                               0.
Pages wired down:                         165275.
Pages purgeable:                           32541.
"Translation faults":                  196873954.
Pages copy-on-write:                     3086841.
Pages zero filled:                     292495234.
Pages reactivated:                       8858569.
Pages purged:                            2515175.
File-backed pages:                        228595.
Anonymous pages:                          396652.
Pages stored in compressor:               413881.
Pages occupied by compressor:             155265.
Decompressions:                          3899051.
Compressions:                            7240140.
Pageins:                                11293234.
Pageouts:                                  5656.
Swapins:                                     68.
Swapouts:                                   368.
Pages tagged:                            129550.
Pages tagged resident:                    89519.
Pages tagged compressed:                  40031.
Pages tag-storage:                        32768.
Pages tag-storage holding tags:            4979.
Pages tag-storage free:                      61.
Pages tag-storage non-tag pageable:       27540.
Pages tag-storage non-tag wired:             73.
Bytes of compressed tags:                6157568.
Tagged compressions:                      438465.
Tagged decompressions:                    365912.
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % df -h
Filesystem        Size    Used   Avail Capacity iused ifree %iused  Mounted on
/dev/disk3s1s1   460Gi    13Gi   320Gi     4%    484k  3.4G    0%   /
devfs            199Ki   199Ki     0Bi   100%     689     0  100%   /dev
/dev/disk3s6     460Gi   1.0Gi   320Gi     1%       1  3.4G    0%   /System/Volumes/VM
/dev/disk3s2     460Gi    11Gi   320Gi     4%    1.1k  3.4G    0%   /System/Volumes/Preboot
/dev/disk3s4     460Gi    22Mi   320Gi     1%      65  3.4G    0%   /System/Volumes/Update
/dev/disk1s2     550Mi   6.0Mi   528Mi     2%       1  5.4M    0%   /System/Volumes/xarts
/dev/disk1s1     550Mi   5.9Mi   528Mi     2%      38  5.4M    0%   /System/Volumes/iSCPreboot
/dev/disk1s3     550Mi   5.6Mi   528Mi     2%     864  5.4M    0%   /System/Volumes/Hardware
/dev/disk3s5     460Gi   114Gi   320Gi    27%    953k  3.4G    0%   /System/Volumes/Data
```

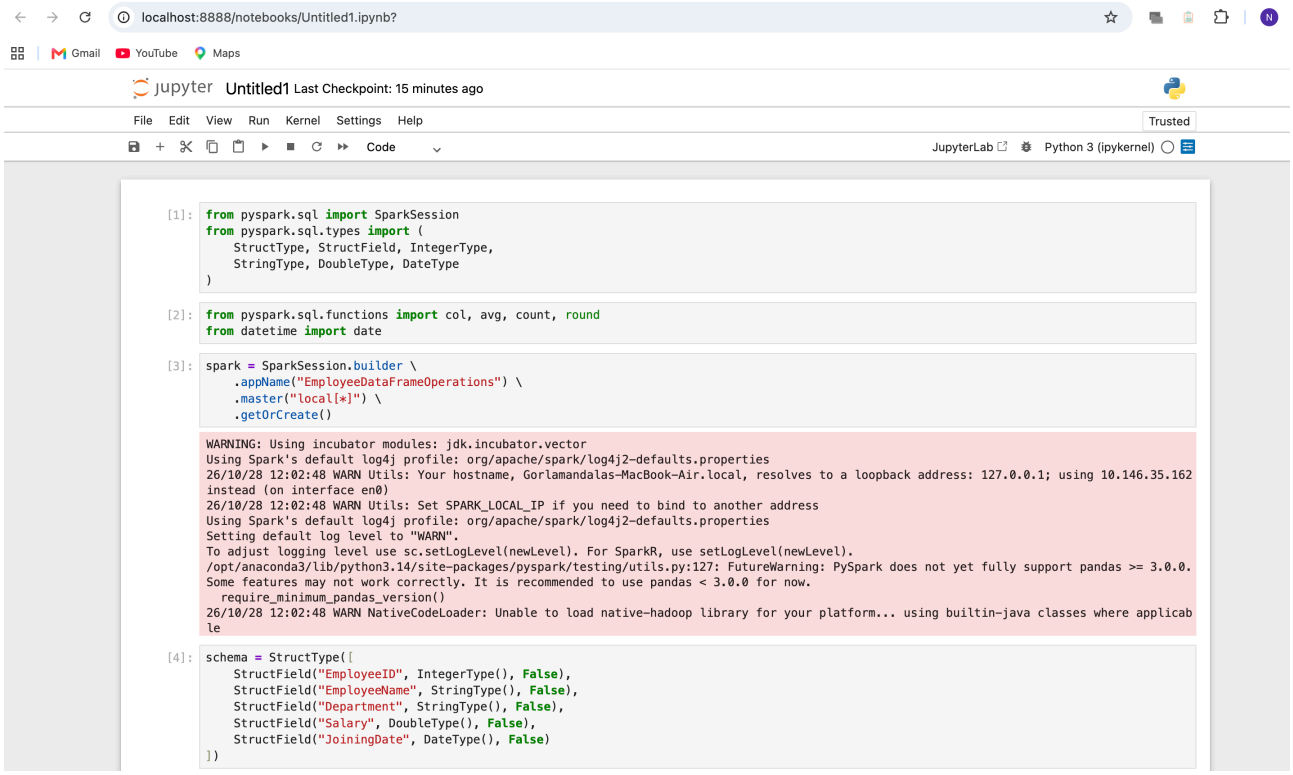


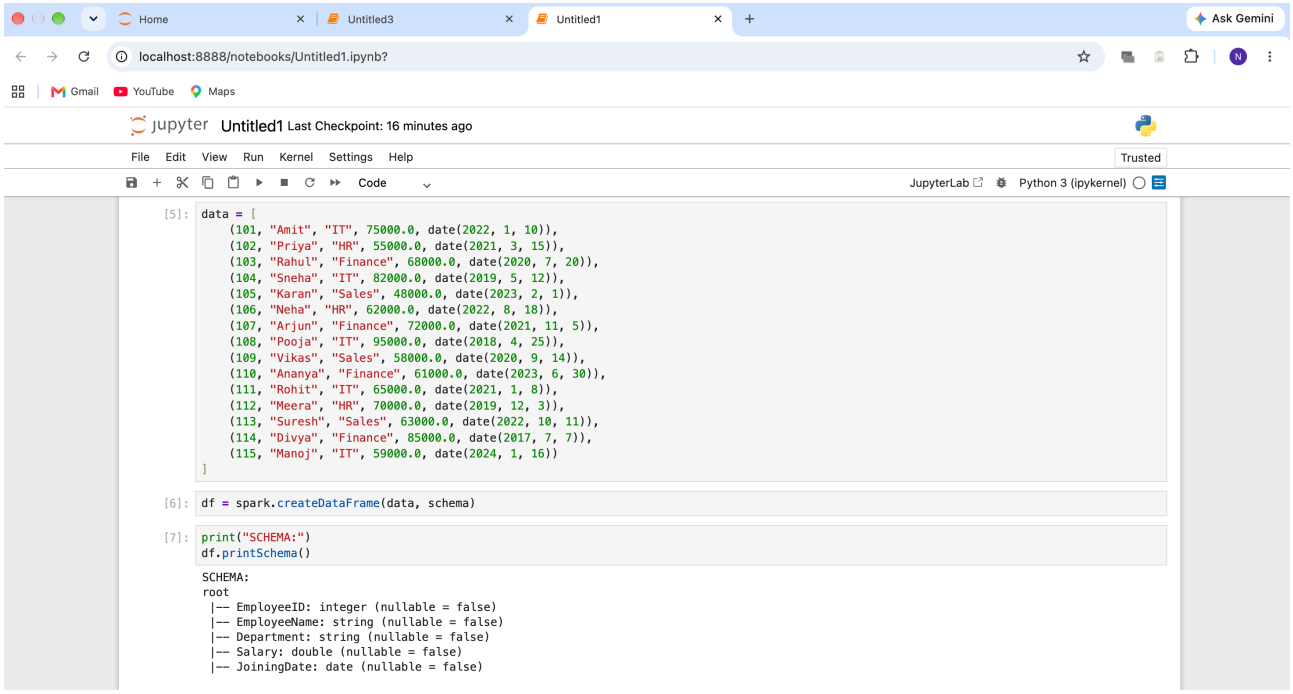
```
Tagged decompressions:                    365912.
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % df -h
Filesystem        Size    Used   Avail Capacity iused ifree %iused  Mounted on
/dev/disk3s1s1   460Gi    13Gi   320Gi     4%    484k  3.4G    0%   /
devfs            199Ki   199Ki     0Bi   100%     689     0  100%   /dev
/dev/disk3s6     460Gi   1.0Gi   320Gi     1%       1  3.4G    0%   /System/Volumes/VM
/dev/disk3s2     460Gi    11Gi   320Gi     4%    1.1k  3.4G    0%   /System/Volumes/Preboot
/dev/disk3s4     460Gi    22Mi   320Gi     1%      65  3.4G    0%   /System/Volumes/Update
/dev/disk1s2     550Mi   6.0Mi   528Mi     2%       1  5.4M    0%   /System/Volumes/xarts
/dev/disk1s1     550Mi   5.9Mi   528Mi     2%      38  5.4M    0%   /System/Volumes/iSCPreboot
/dev/disk1s3     550Mi   5.6Mi   528Mi     2%     864  5.4M    0%   /System/Volumes/Hardware
/dev/disk3s5     460Gi   114Gi   320Gi    27%    953k  3.4G    0%   /System/Volumes/Data
map auto_home      0Bi     0Bi     0Bi   100%       0     0     -   /System/Volumes/Data/home
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % df -h
Filesystem        Size    Used   Avail Capacity iused ifree %iused  Mounted on
/dev/disk3s1s1   460Gi    13Gi   320Gi     4%    484k  3.4G    0%   /
devfs            199Ki   199Ki     0Bi   100%     689     0  100%   /dev
/dev/disk3s6     460Gi   1.0Gi   320Gi     1%       1  3.4G    0%   /System/Volumes/VM
/dev/disk3s2     460Gi    11Gi   320Gi     4%    1.1k  3.4G    0%   /System/Volumes/Preboot
/dev/disk3s4     460Gi    22Mi   320Gi     1%      65  3.4G    0%   /System/Volumes/Update
/dev/disk1s2     550Mi   6.0Mi   528Mi     2%       1  5.4M    0%   /System/Volumes/xarts
/dev/disk1s1     550Mi   5.9Mi   528Mi     2%      38  5.4M    0%   /System/Volumes/iSCPreboot
/dev/disk1s3     550Mi   5.6Mi   528Mi     2%     864  5.4M    0%   /System/Volumes/Hardware
/dev/disk3s5     460Gi   114Gi   320Gi    27%    953k  3.4G    0%   /System/Volumes/Data
map auto_home      0Bi     0Bi     0Bi   100%       0     0     -   /System/Volumes/Data/home
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % mkdir -p ~/finance_assignme
nt/application
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % find ~/finance_assignment/a
pplication -type f -size +100M
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment % find /opt/myapp -type f -si
ze +100M
find: /opt/myapp: No such file or directory
(base) gorlamandalanandasai@Gorlamandalas-MacBook-Air finance_assignment %
```

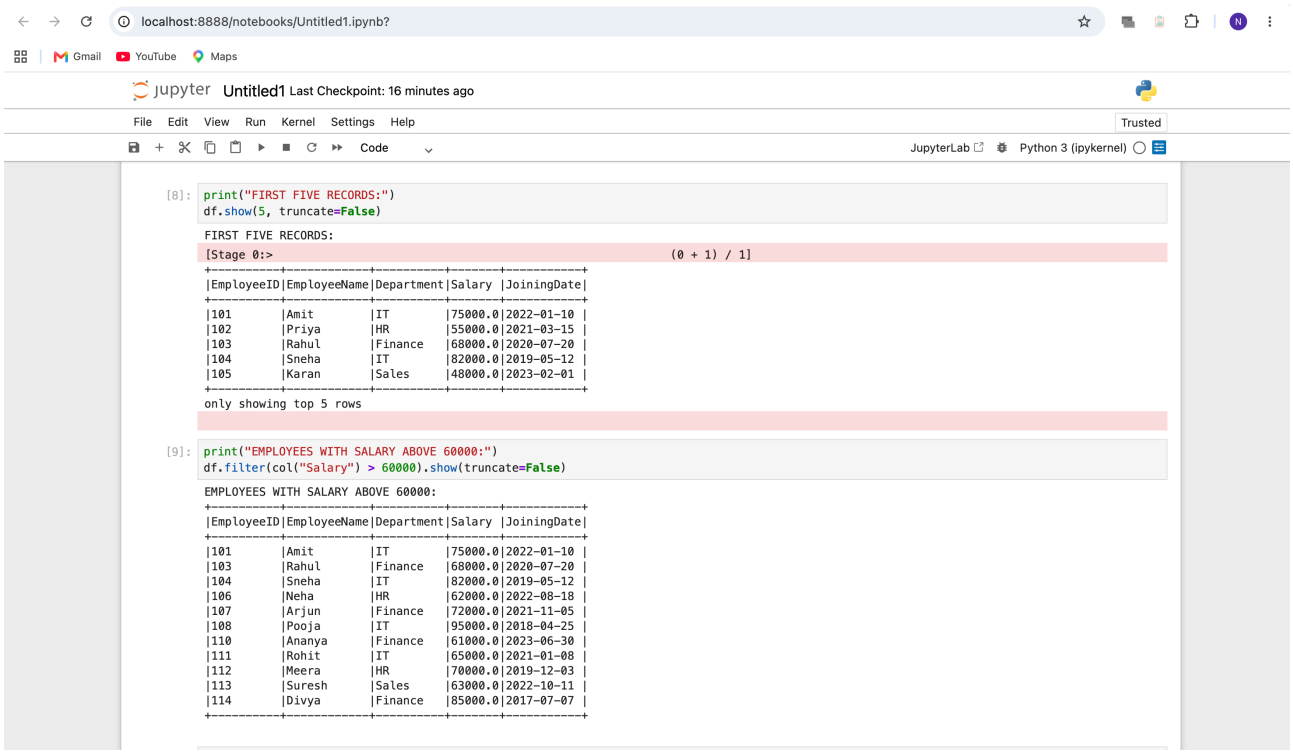
Bash Scripting:

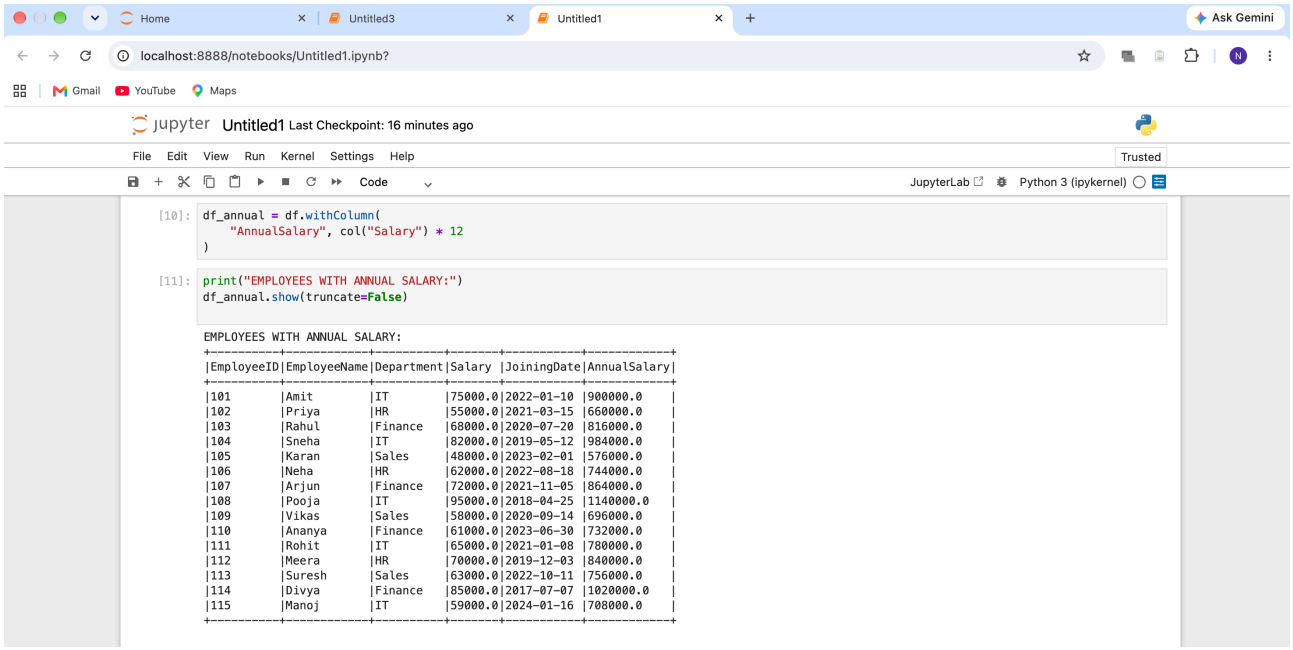


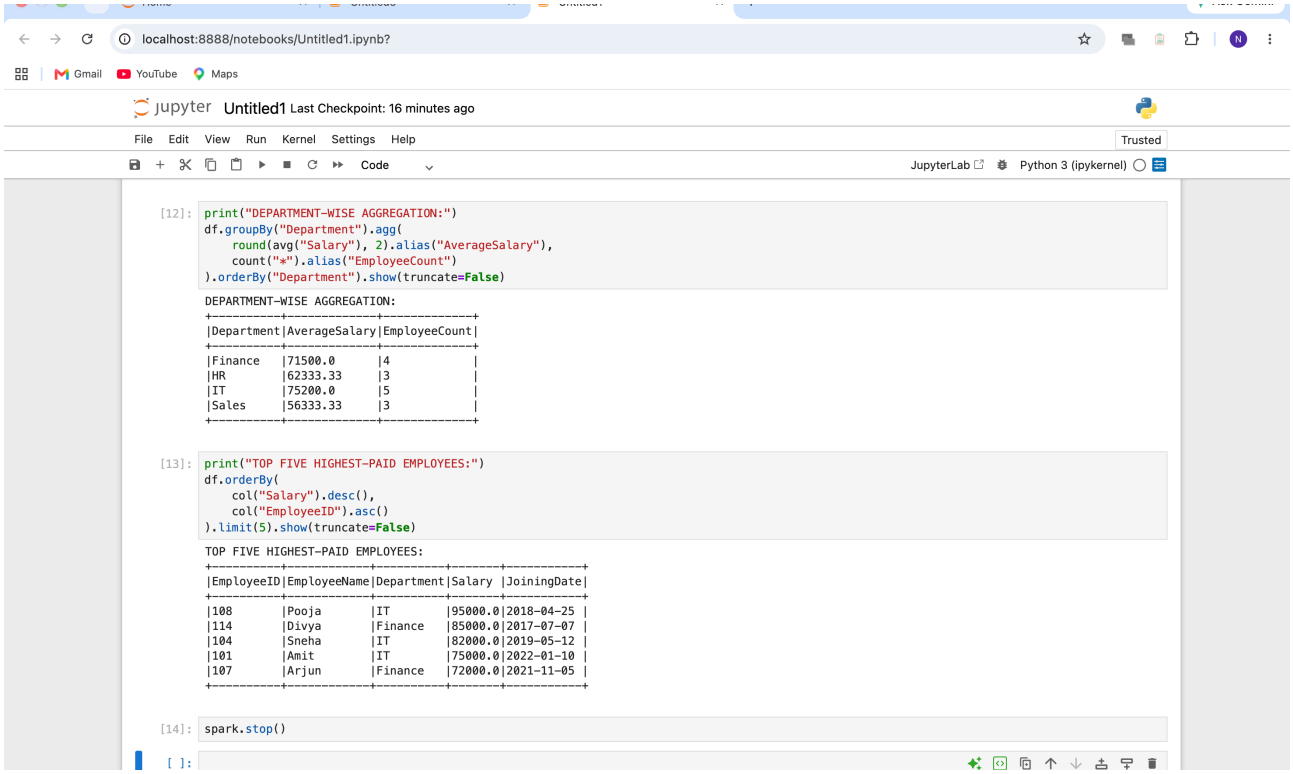
Question 7: PySpark DataFrame Operations &
Question 8: Data Cleaning and Aggregation

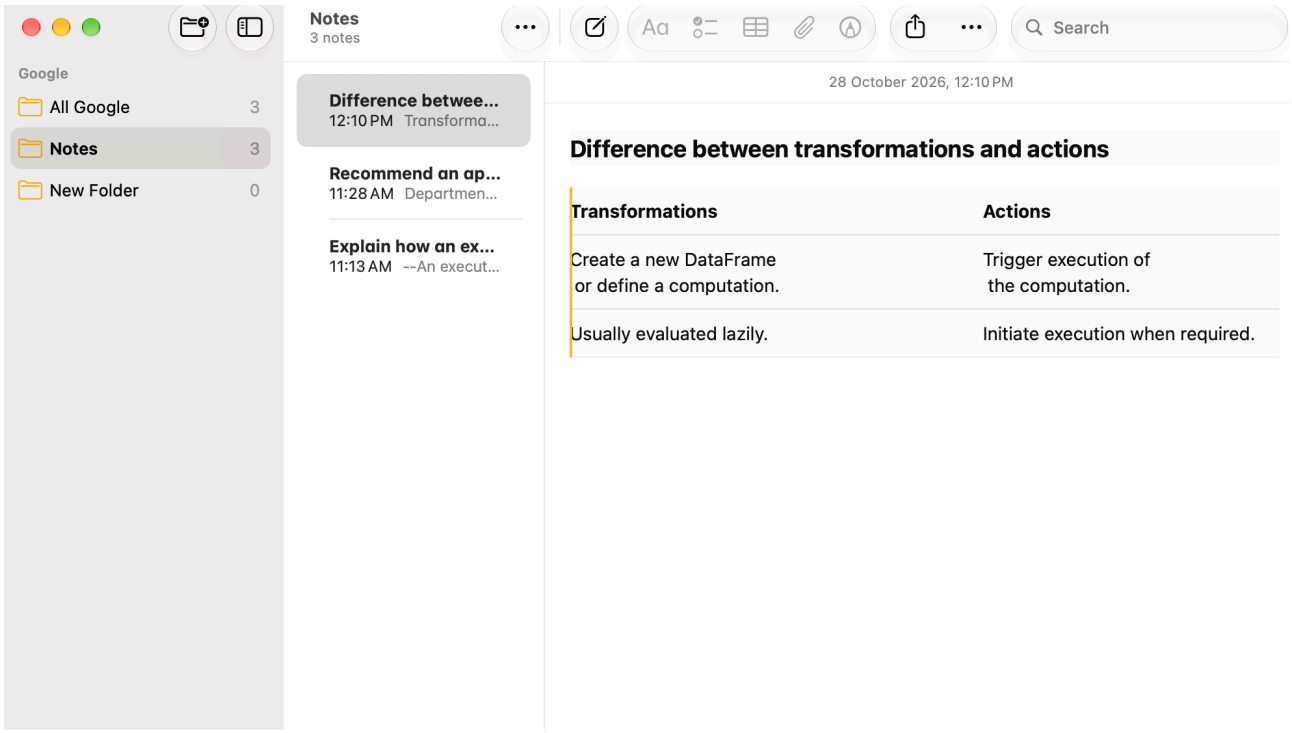





localhost:8888/notebooks/Untitled1.ipynb?
Gmail
YouTube
Maps
jupyter Untitled1 Last Checkpoint: 16 minutes ago
File Edit View Run Kernel Settings Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
[8]:
print("FIRST FIVE RECORDS:")
df.show(5, truncate=False)
FIRST FIVE RECORDS:
[Stage 0:> (0 + 1) / 1]
+----------+------------+----------+-------+-----------+
|EmployeeID|EmployeeName|Department|Salary |JoiningDate|
+----------+------------+----------+-------+-----------+
101	Amit	IT	75000.0	2022-01-10
102	Priya	HR	55000.0	2021-03-15
103	Rahul	Finance	68000.0	2020-07-20
104	Sneha	IT	82000.0	2019-05-12
105	Karan	Sales	48000.0	2023-02-01
+----------+------------+----------+-------+-----------+
only showing top 5 rows
[9]:
print("EMPLOYEES WITH SALARY ABOVE 60000:")
df.filter(col("Salary") > 60000).show(truncate=False)
EMPLOYEES WITH SALARY ABOVE 60000:
+----------+------------+----------+-------+-----------+
|EmployeeID|EmployeeName|Department|Salary |JoiningDate|
+----------+------------+----------+-------+-----------+
101	Amit	IT	75000.0	2022-01-10
103	Rahul	Finance	68000.0	2020-07-20
104	Sneha	IT	82000.0	2019-05-12
106	Neha	HR	62000.0	2022-08-18
107	Arjun	Finance	72000.0	2021-11-05
108	Pooja	IT	95000.0	2018-04-25
110	Ananya	Finance	61000.0	2023-06-30
111	Rohit	IT	65000.0	2021-01-08
112	Meera	HR	70000.0	2019-12-03
113	Suresh	Sales	63000.0	2022-10-11
114	Divya	Finance	85000.0	2017-07-07
+----------+------------+----------+-------+-----------+


Home
Untitled3
Untitled1
Ask Gemini
localhost:8888/notebooks/Untitled1.ipynb?
Gmail
YouTube
Maps
jupyter Untitled1 Last Checkpoint: 16 minutes ago
File Edit View Run Kernel Settings Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
[10]:
df_annual = df.withColumn(
"AnnualSalary", col("Salary") * 12
)
[11]:
print("EMPLOYEES WITH ANNUAL SALARY:")
df_annual.show(truncate=False)
EMPLOYEES WITH ANNUAL SALARY:
+----------+------------+----------+-------+-----------+-------------+
|EmployeeID|EmployeeName|Department|Salary |JoiningDate|AnnualSalary|
+----------+------------+----------+-------+-----------+-------------+
101	Amit	IT	75000.0	2022-01-10	900000.0
102	Priya	HR	55000.0	2021-03-15	660000.0
103	Rahul	Finance	68000.0	2020-07-20	816000.0
104	Sneha	IT	82000.0	2019-05-12	984000.0
105	Karan	Sales	48000.0	2023-02-01	576000.0
106	Neha	HR	62000.0	2022-08-18	744000.0
107	Arjun	Finance	72000.0	2021-11-05	864000.0
108	Pooja	IT	95000.0	2018-04-25	1140000.0
109	Vikas	Sales	58000.0	2020-09-14	696000.0
110	Ananya	Finance	61000.0	2023-06-30	732000.0
111	Rohit	IT	65000.0	2021-01-08	780000.0
112	Meera	HR	70000.0	2019-12-03	840000.0
113	Suresh	Sales	63000.0	2022-10-11	756000.0
114	Divya	Finance	85000.0	2017-07-07	1020000.0
115	Manoj	IT	59000.0	2024-01-16	708000.0
+----------+------------+----------+-------+-----------+-------------+

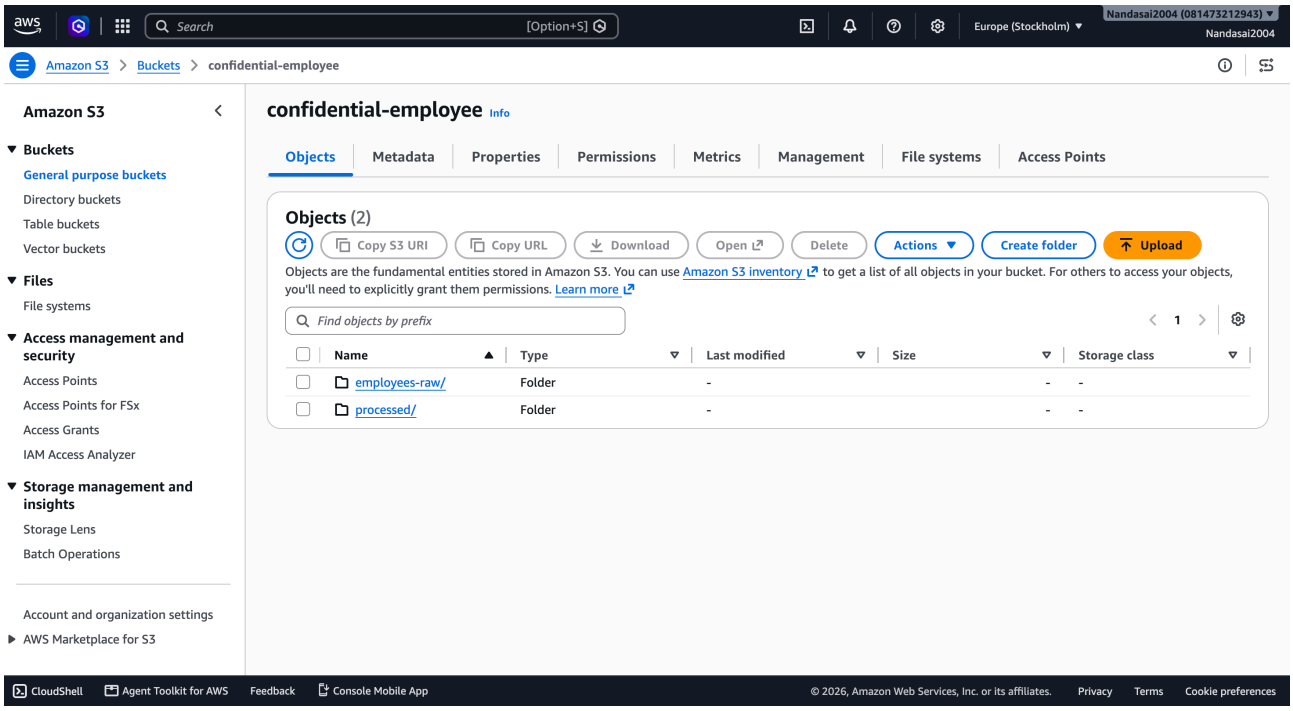

localhost:8888/notebooks/Untitled1.ipynb?
Gmail
YouTube
Maps
jupyter Untitled1 Last Checkpoint: 16 minutes ago
File Edit View Run Kernel Settings Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
[12]:
print("DEPARTMENT-WISE AGGREGATION:")
df.groupBy("Department").agg(
round(avg("Salary"), 2).alias("AverageSalary"),
count("*").alias("EmployeeCount")
).orderBy("Department").show(truncate=False)
DEPARTMENT-WISE AGGREGATION:
+----------+-------------+-------------+
|Department|AverageSalary|EmployeeCount|
+----------+-------------+-------------+
Finance	71500.0	4
HR	62333.33	3
IT	75200.0	5
Sales	56333.33	3
+----------+-------------+-------------+
[13]:
print("TOP FIVE HIGHEST-PAID EMPLOYEES:")
df.orderBy(
col("Salary").desc(),
col("EmployeeID").asc()
).limit(5).show(truncate=False)
TOP FIVE HIGHEST-PAID EMPLOYEES:
+----------+------------+----------+-------+-----------+
|EmployeeID|EmployeeName|Department|Salary |JoiningDate|
+----------+------------+----------+-------+-----------+
108	Pooja	IT	95000.0	2018-04-25
114	Divya	Finance	85000.0	2017-07-07
104	Sneha	IT	82000.0	2019-05-12
101	Amit	IT	75000.0	2022-01-10
107	Arjun	Finance	72000.0	2021-11-05
+----------+------------+----------+-------+-----------+
[14]:
spark.stop()
[ ]:

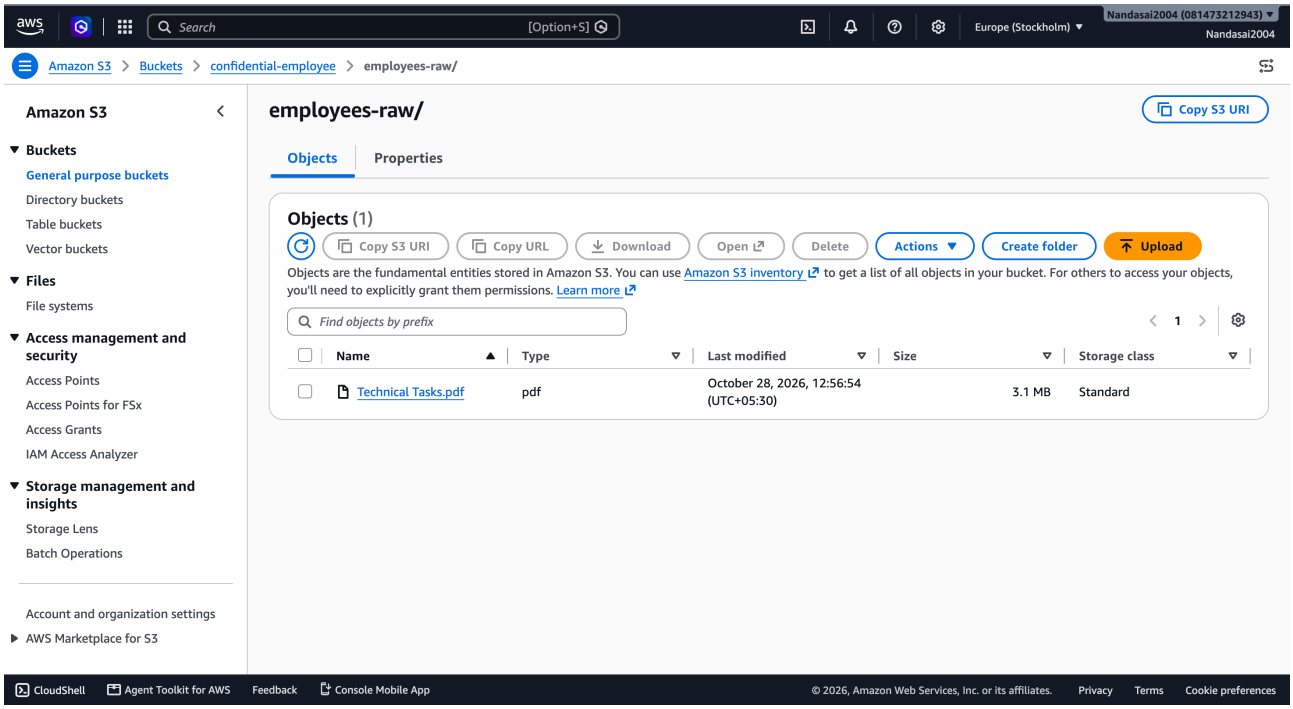

Notes
3 notes
Search
Google
All Google 3
Notes 3
New Folder 0
Difference betwee...
12:10 PM Transforma...
Recommend an ap...
11:28 AM Departmen...
Explain how an ex...
11:13 AM --An execut...
28 October 2026, 12:10 PM
Difference between transformations and actions
Transformations
Actions
Create a new DataFrame or define a computation.
Trigger execution of the computation.
Usually evaluated lazily.
Initiate execution when required.

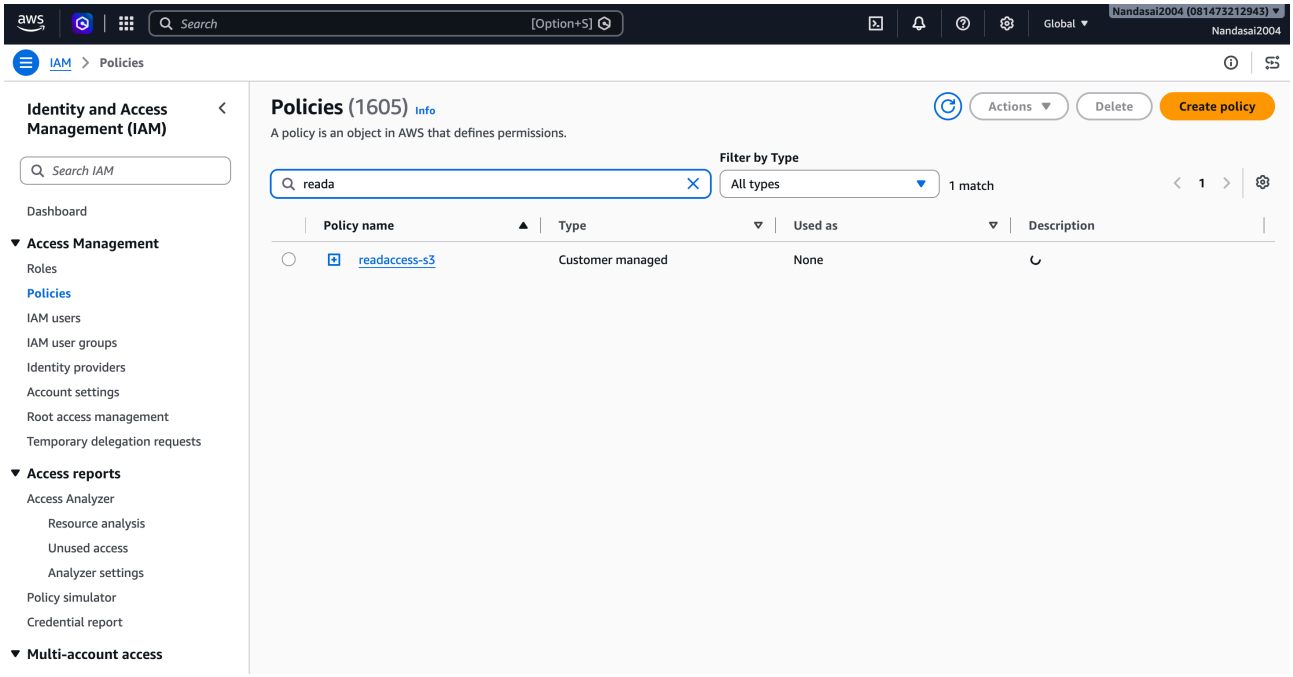
SECTION D: AWS CLOUD SERVICES

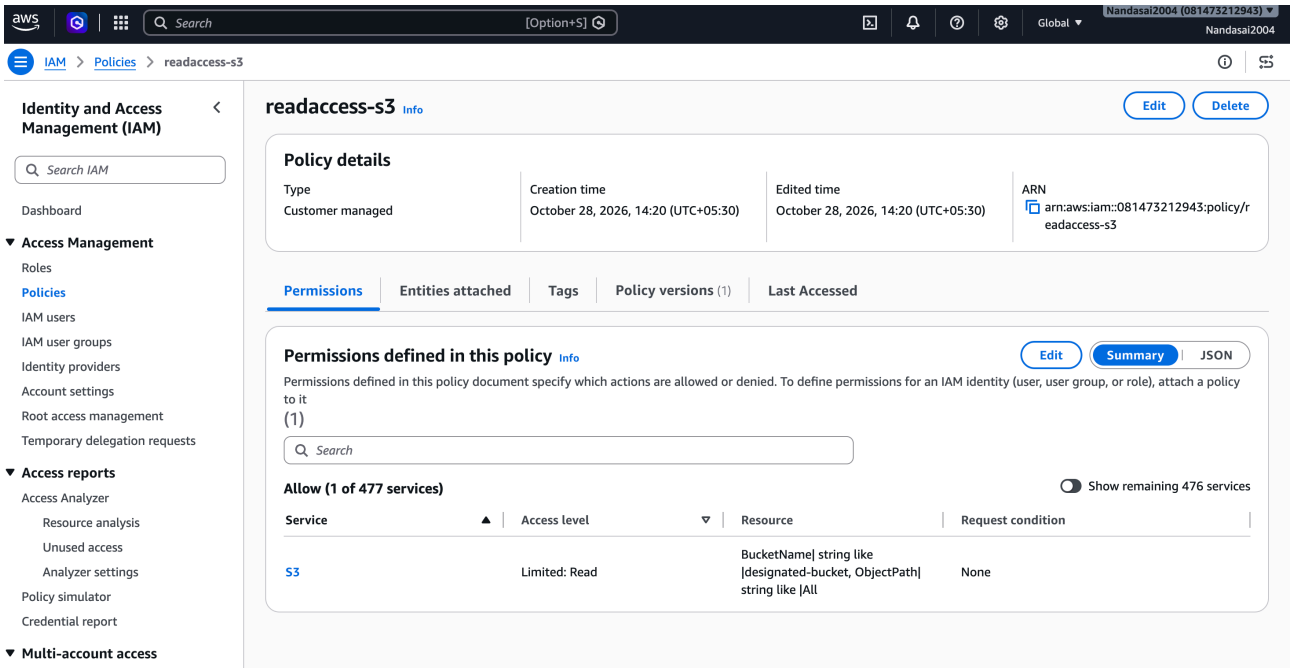
Question 10: S3 and IAM Security

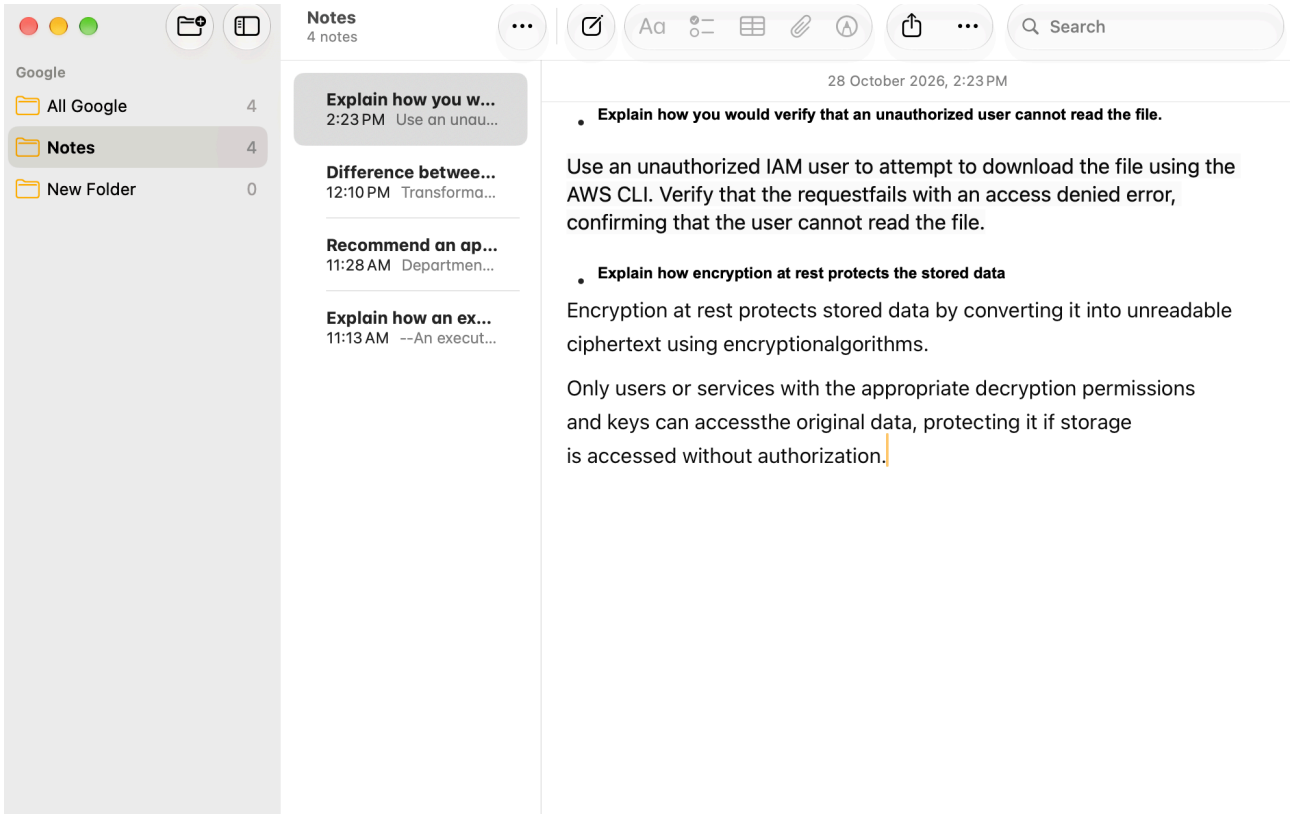
Create an S3 bucket with Block Public Access enabled and upload a sample file



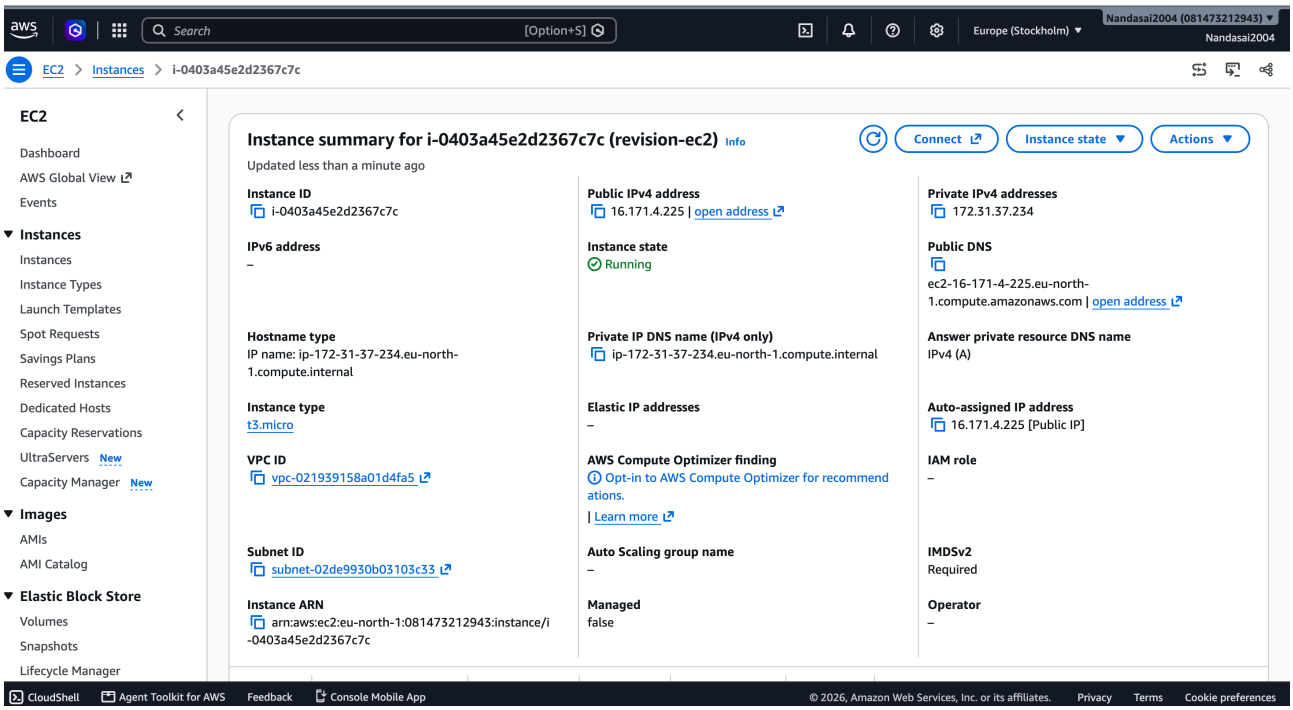


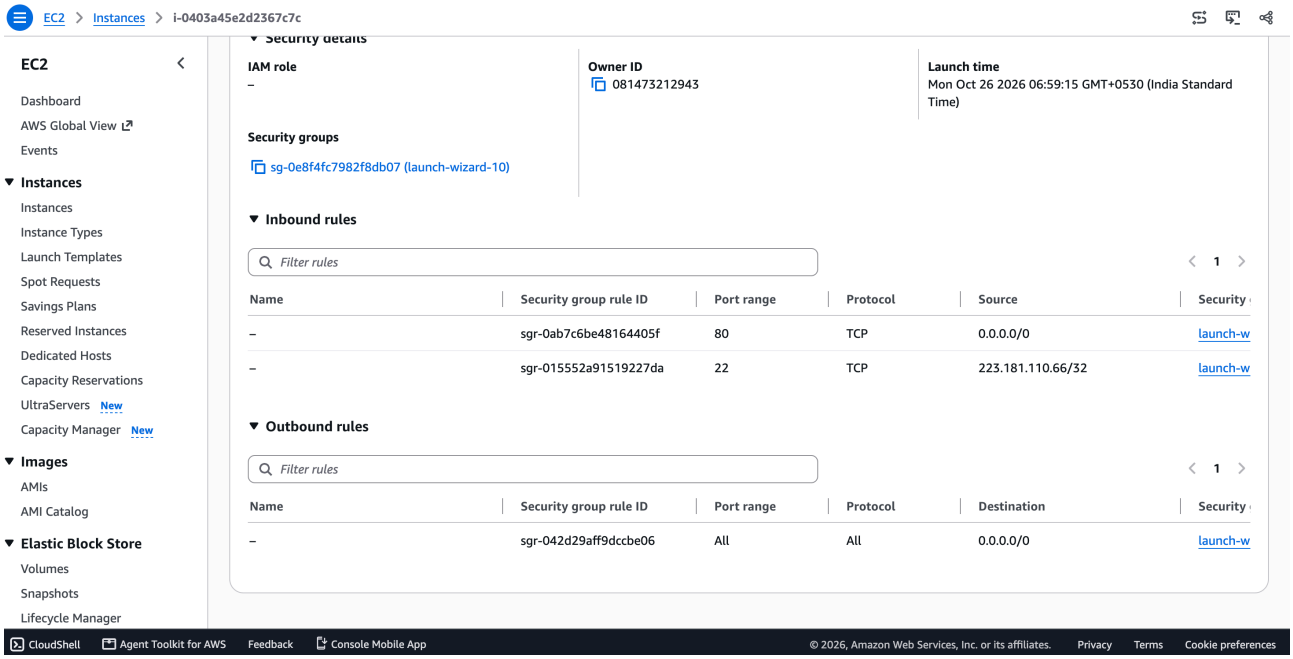

Search
[Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Policies
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
Policies (1605) Info
A policy is an object in AWS that defines permissions.
Actions
Delete
Create policy
Filter by Type
reada
All types
1 match
1
Policy name
Type
Used as
Description
readaccess-s3
Customer managed
None


Search
[Option+S]
Global
Nandasai2004 (081473212943)
Nandasai2004
IAM > Policies > readaccess-s3
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
readaccess-s3 Info
Edit
Delete
Policy details
Type
Customer managed
Creation time
October 28, 2026, 14:20 (UTC+05:30)
Edited time
October 28, 2026, 14:20 (UTC+05:30)
ARN
arn:aws:iam::081473212943:policy/readaccess-s3
Permissions
Entities attached
Tags
Policy versions (1)
Last Accessed
Permissions defined in this policy Info
Edit
Summary
JSON
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it
(1)
Search
Allow (1 of 477 services)
Show remaining 476 services
Service
Access level
Resource
Request condition
S3
Limited: Read
BucketName| string like |designated-bucket, ObjectPath| string like |All
None


Notes
4 notes
Search
Google
All Google 4
Notes 4
New Folder 0
Explain how you w...
2:23 PM Use an unau...
Difference betwee...
12:10 PM Transforma...
Recommend an ap...
11:28 AM Departmen...
Explain how an ex...
11:13 AM --An execut...
28 October 2026, 2:23 PM
Explain how you would verify that an unauthorized user cannot read the file.
Use an unauthorized IAM user to attempt to download the file using the AWS CLI. Verify that the requestfails with an access denied error, confirming that the user cannot read the file.
Explain how encryption at rest protects the stored data
Encryption at rest protects stored data by converting it into unreadable ciphertext using encryptionalgorithms.
Only users or services with the appropriate decryption permissions and keys can accessthe original data, protecting it if storage is accessed without authorization.

Question 11: EC2 Troubleshooting and CloudWatch





Notes
5 notes

Search

Google
All Google 5
Notes 5
New Folder 0

Explain how secur...
2:45 PM Security Gro...

Explain how you w...
2:23 PM Use an unau...

Difference betwee...
12:10 PM Transforma...

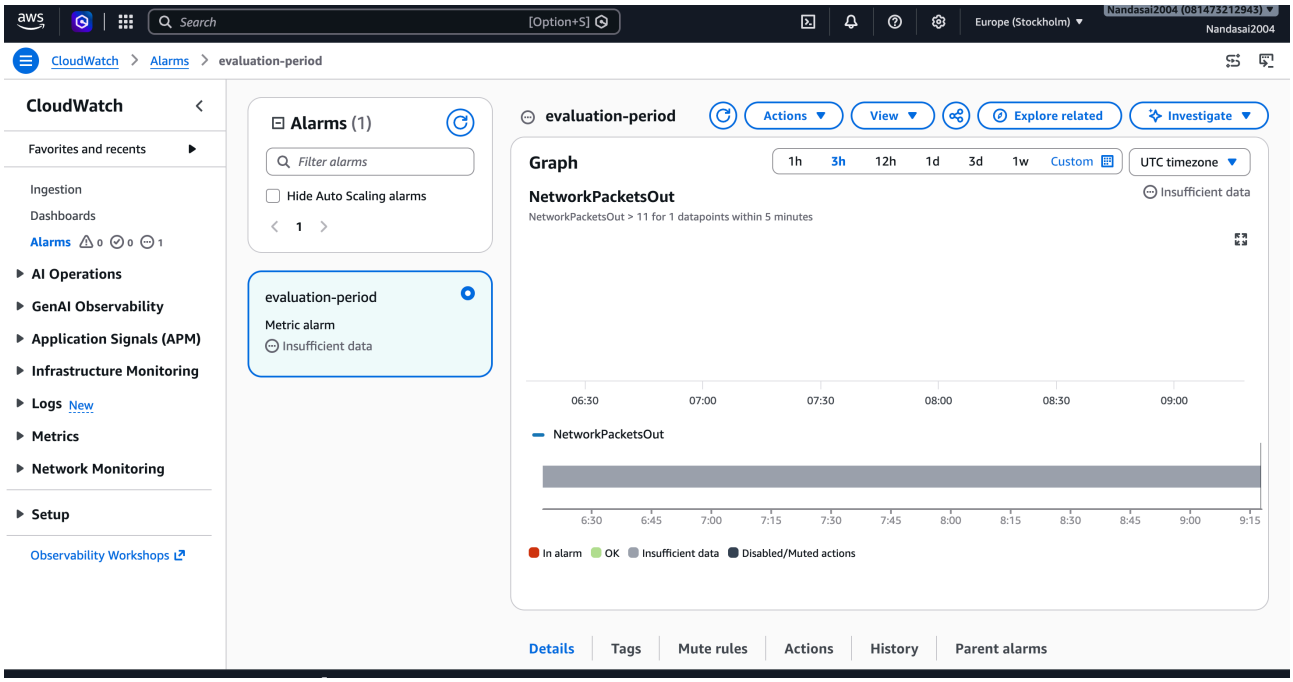
Recommend an ap...
11:28 AM Departmen...

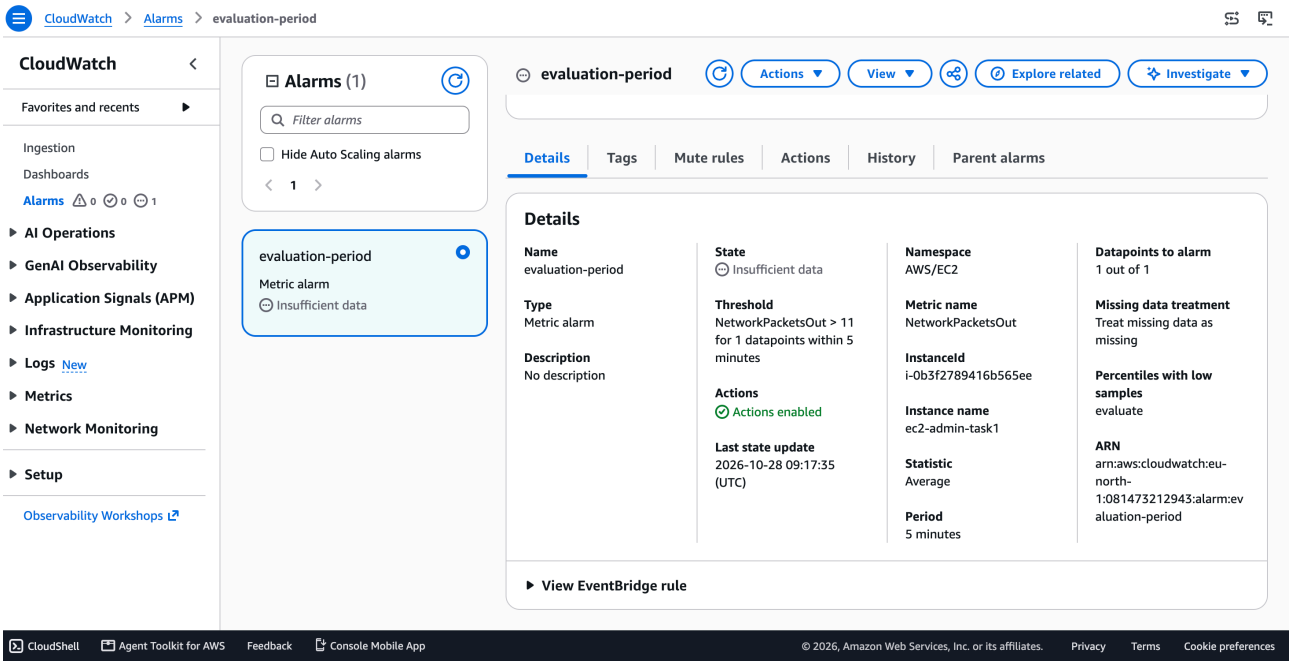
Explain how an ex...
11:13 AM --An execut...

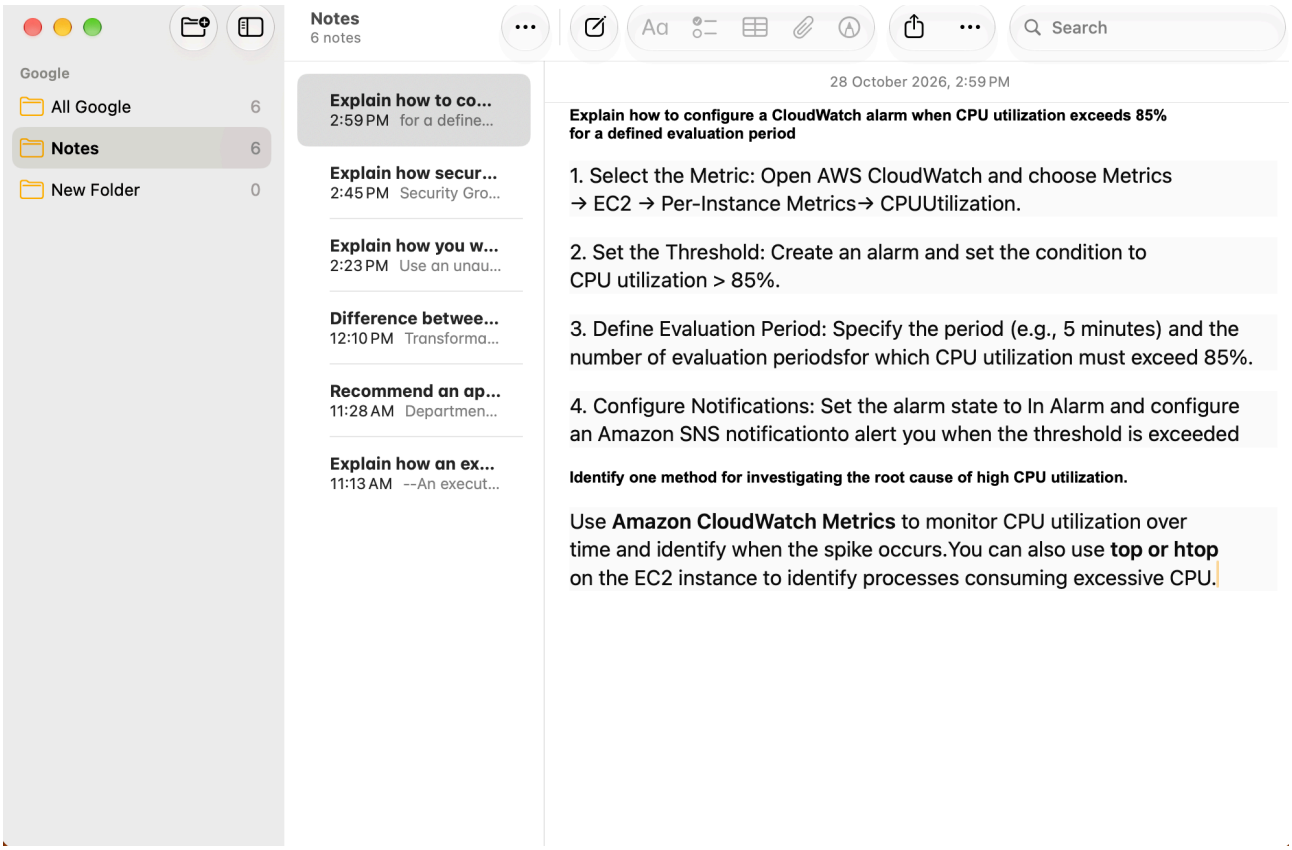
28 October 2026, 2:45 PM

Explain how security groups and route tables can affect application connectivity

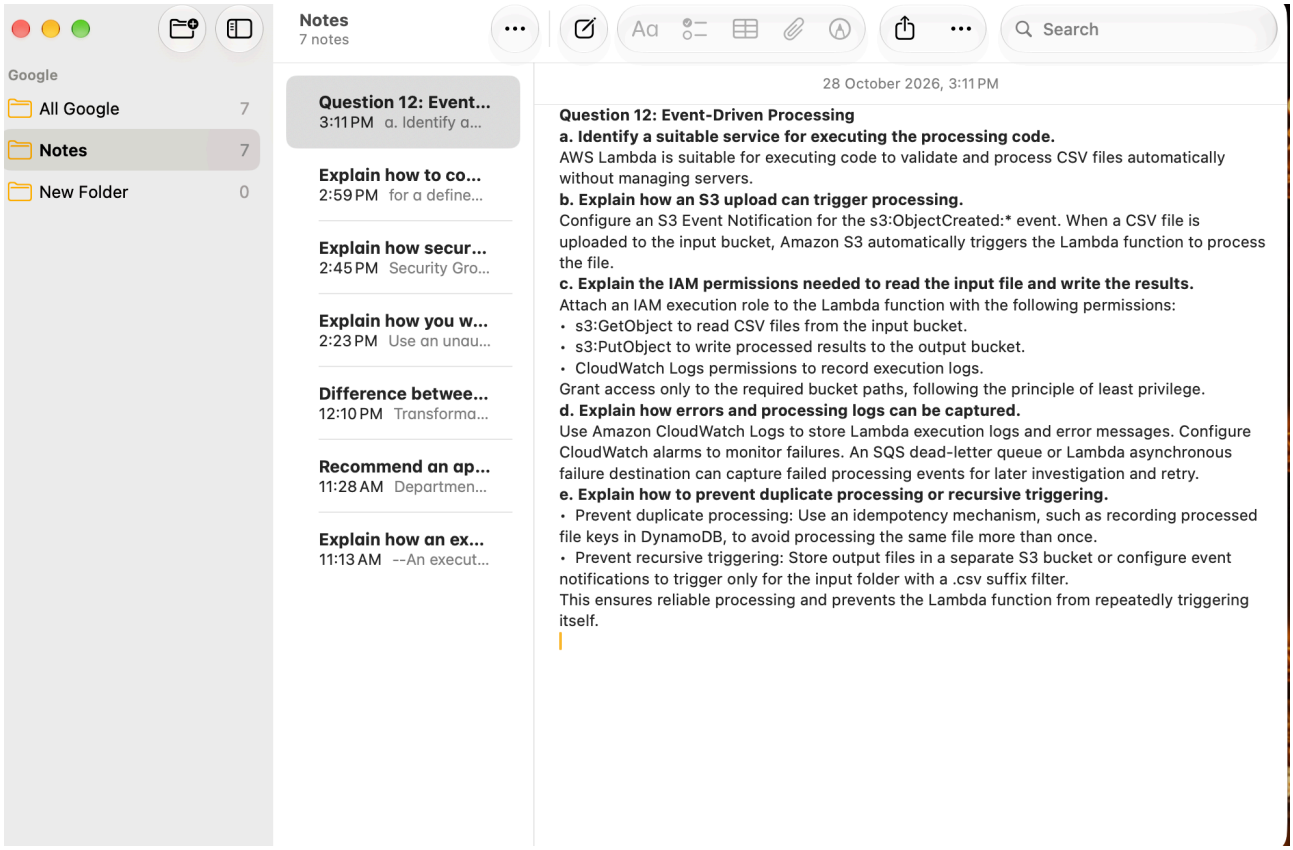
- **Security Groups:** These act as virtual firewalls. If inbound rules do not allow the required ports(e.g., HTTP port 80 or HTTPS port 443), users cannot access the application.
- **Route Tables:** These determine where network traffic is directed. If a public subnet lacks a route to anInternet Gateway, or routes are incorrectly configured, users may not be able to reach the application.



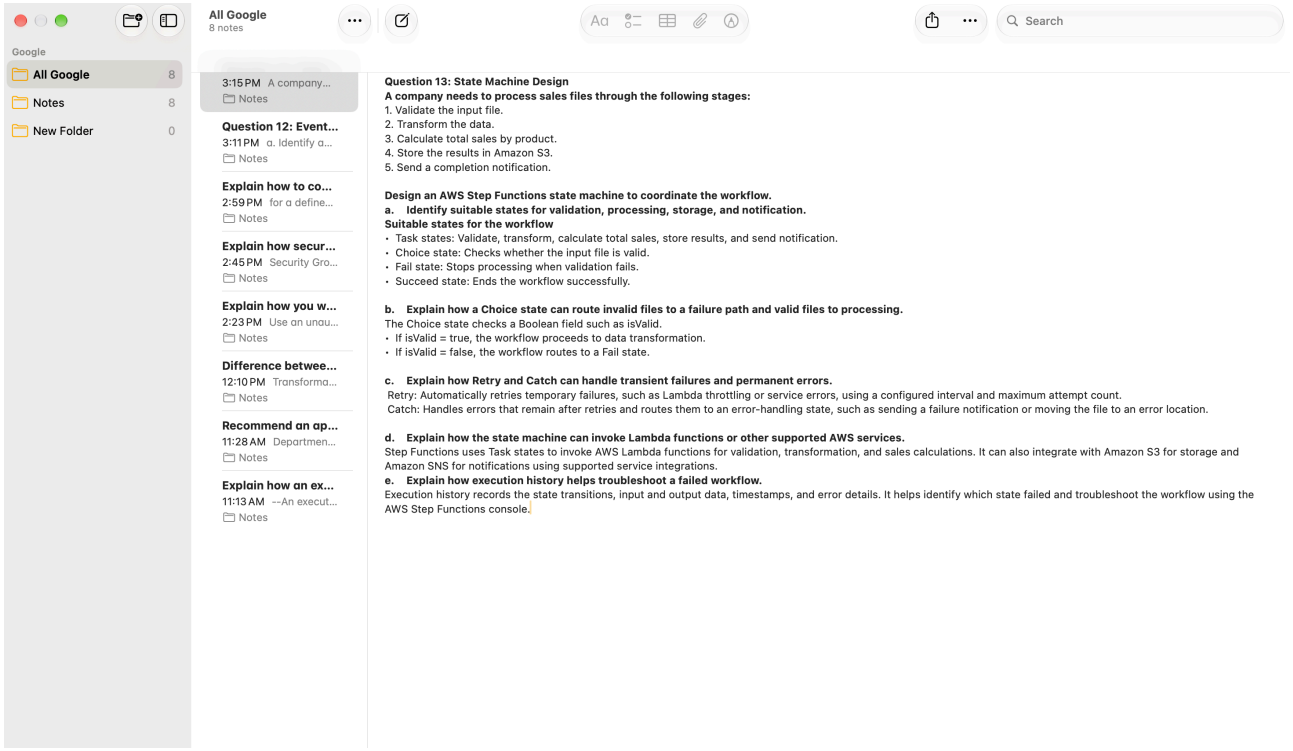

CloudWatch > Alarms > evaluation-period
CloudWatch
Favorites and recents
Ingestion
Dashboards
Alarms 0 0 1
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs New
Metrics
Network Monitoring
Setup
Observability Workshops
Alarms (1)
Filter alarms
Hide Auto Scaling alarms
1
evaluation-period
Metric alarm
Insufficient data
evaluation-period
Actions
View
Explore related
Investigate
Details
Tags
Mute rules
Actions
History
Parent alarms
Details
Name
evaluation-period
Type
Metric alarm
Description
No description
State
Insufficient data
Threshold
NetworkPacketsOut > 11 for 1 datapoints within 5 minutes
Actions
Actions enabled
Last state update
2026-10-28 09:17:35 (UTC)
Namespace
AWS/EC2
Metric name
NetworkPacketsOut
InstanceId
i-0b3f2789416b565ee
Instance name
ec2-admin-task1
Statistic
Average
Period
5 minutes
Datapoints to alarm
1 out of 1
Missing data treatment
Treat missing data as missing
Percentiles with low samples
evaluate
ARN
arn:aws:cloudwatch:eu-north-1:081473212943:alarm:evaluation-period
View EventBridge rule
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Notes
6 notes
Search
Google
All Google 6
Notes 6
New Folder 0
Explain how to co...
2:59 PM for a define...
Explain how secur...
2:45 PM Security Gro...
Explain how you w...
2:23 PM Use an unau...
Difference betwee...
12:10 PM Transforma...
Recommend an ap...
11:28 AM Departmen...
Explain how an ex...
11:13 AM --An execut...
28 October 2026, 2:59 PM
Explain how to configure a CloudWatch alarm when CPU utilization exceeds 85% for a defined evaluation period
1. Select the Metric: Open AWS CloudWatch and choose Metrics → EC2 → Per-Instance Metrics→ CPUUtilization.
2. Set the Threshold: Create an alarm and set the condition to CPU utilization > 85%.
3. Define Evaluation Period: Specify the period (e.g., 5 minutes) and the number of evaluation periodsfor which CPU utilization must exceed 85%.
4. Configure Notifications: Set the alarm state to In Alarm and configure an Amazon SNS notificationto alert you when the threshold is exceeded
Identify one method for investigating the root cause of high CPU utilization.
Use Amazon CloudWatch Metrics to monitor CPU utilization over time and identify when the spike occurs.You can also use top or htop on the EC2 instance to identify processes consuming excessive CPU.

Question 12: Event-Driven Processing



Question 13: State Machine Design



Question 14: Step Functions Implementation

Write an ASL JSON definition containing the three sequential tasks. Use placeholders for resource ARNs where necessary.

```
{
  "StartAt": "ValidateFile",
  "States": {
    "ValidateFile": {
      "Type": "Task",
      "Resource": "arn:aws:lambda:REGION:ACCOUNT:function:ValidateFile",
      "Next": "TransformData"
    },
    "TransformData": {
      "Type": "Task",
      "Resource": "arn:aws:lambda:REGION:ACCOUNT:function:TransformData",
      "Next": "LoadResults"
    },
    "LoadResults": {
      "Type": "Task",
      "Resource": "arn:aws:lambda:REGION:ACCOUNT:function:LoadResults",
      "End": true
    }
  }
}
```

`ValidateFile` runs first.
Then `TransformData` runs.
Finally `LoadResults` runs and the workflow ends

b. Retry Policy for Transformation

```
"Retry": [
  {
    "ErrorEquals": ["States.TaskFailed"],
    "IntervalSeconds": 5,
    "MaxAttempts": 3
  }
]
```

Retry when a temporary task failure occurs.
Wait 5 seconds before retrying.
Try a maximum of 3 times.

c. Catch for Loading Failure

```
"Catch": [
  {
    "ErrorEquals": ["States.ALL"],
    "Next": "HandleError"
  }
]
```

1. Catch errors from `LoadResults`.
2. Move to `HandleError`.
3. `HandleError` records/handles the failure.

d. Passing Data

1. Each task produces an **output**.
2. That output becomes the **input** of the next task.
3. Example: `ValidateFile → TransformData → LoadResults`.

e. Testing

1. Give a **valid file** → all three tasks should succeed.
2. Give an **invalid file** → validation should fail and workflow should stop.
3. Simulate failures → transformation should retry and loading failure should go to `HandleError`.

Question 15: Azure Data Lake Storage and Security

Question 15: Azure Data Lake Storage and Security

a. Configure ADLS Gen2 Storage Account

1. Go to Azure Portal → Storage Accounts → Create.
2. Select Data Lake Storage Gen2 and enable Hierarchical Namespace.
3. Create the storage account.

b. Create Filesystem and Upload Dataset

1. Open the storage account → Data storage → Containers.
2. Create a filesystem/container, for example `business-data`.
3. Upload a sample file such as `employees.csv`.

c. Microsoft Entra ID and RBAC

1. Microsoft Entra ID identifies and authenticates users.
2. Use Azure RBAC to assign roles such as **Storage Blob Data Reader** to authorized users.
3. Give users only the permissions they need (**least privilege**), while unauthorized users receive no data-access role.

d. Verify Unauthorized Access

1. Log in as a user who has **no permission** to the storage data.
2. Try to open or download the dataset.
3. Access should be **denied**, confirming that RBAC is working correctly.

Question 16: Azure VM Troubleshooting

Question 16: Azure VM Troubleshooting

a. Verify VM provisioning and power states

1. Go to **Azure Portal → Virtual Machines → Select VM → Overview**.
2. Check **Power state** — it should be **Running**.
3. Check **Provisioning state** — it should be **Succeeded**.

---

b. Network Security Group (NSG)

1. NSG controls **inbound and outbound network traffic**.
2. If the required port, such as **SSH (22)** or **RDP (3389)**, is blocked, administrators cannot connect.
3. Check the NSG rules and allow only the required traffic from trusted IP addresses.

---

c. Subnet and Routing

1. The VM must be connected to the **correct subnet and virtual network**.
2. Incorrect or missing **route table/route** entries can prevent traffic from reaching the VM.
3. Check the subnet, route table, and next-hop configuration.

---

d. Secure Troubleshooting Method

Use **Azure Bastion** instead of exposing SSH/RDP directly to the public internet.

1. Azure Bastion provides secure browser-based access to the VM through the Azure Portal.
2. It avoids giving the VM a public IP and avoids opening SSH/RDP ports to the entire internet.

Simple flow:

```
Administrator
      ↓
Azure Portal
      ↓
Azure Bastion
      ↓
Private VM
```

Question 17: Azure Data Integration

Question 17: Azure Data Integration

a. Service to orchestrate data movement

Use **Azure Data Factory (ADF)**.

1. ADF creates and manages data pipelines.
2. It can automatically move data from one location to another.
3. It supports CSV files and Azure Data Lake Storage Gen2.

---

b. Copy CSV to ADLS Gen2

1. Create an **ADF pipeline** with a **Copy Data** activity.
2. Configure the CSV file as the **source**.
3. Configure **ADLS Gen2** as the destination and run the pipeline.

```
CSV File → Azure Data Factory → ADLS Gen2
```

---

c. Organize files by ingestion date — 1 Mark

Create folders using the ingestion date.

Example:

```
transactions/
├── 2026/
│   ├── 10/
│   │   ├── 28/
│   │   │   └── transactions.csv
```

ADF can use **dynamic expressions** to automatically create the date-based folder.

d. Handle malformed records and missing values

1. Use **ADF Mapping Data Flow** to validate and transform the data.
2. Send malformed records to a separate **error/quarantine folder**.
3. Replace missing values with suitable defaults or remove the records based on business rules.

```
Valid Records → Processed Data
Invalid Records → Error/Quarantine Folder
```

---

e. Monitor pipeline and validate completeness

1. Use **ADF Monitor** to check pipeline status, failures, and execution time.
2. Check the number of records/files processed.
3. Compare the **source record count with the destination record count** to verify data completeness.